

AMG Wind Farm

Power Forecasting System

Technical Project Report

Project	Multi-Horizon Wind Power Forecasting
Farm Capacity	26.4 MW (12 x 2,200 kW Turbines)
Raw Data Period	2021-01-01 to 2026-12-07
Models	XGBoost + LightGBM (130 ML models, 520 artifacts)
Horizons	10 min, 30 min, 1 h, 6 h, 24 h
API Framework	FastAPI + Uvicorn (28 endpoints)
Report Date	2026-08-01
Generated	August 03, 2026 at 10:49
Git Commit	f881ff0
Version	2.1.0

Table of Contents

- 1. Executive Summary 4
- 2. Project Overview 6
 - 2.1 Wind Farm Description 6
 - 2.2 Project Objectives 6
- 3. Data Description 7
 - 3.1 SCADA Data Overview 7
 - 3.2 Data Quality Analysis 8
 - 3.3 Turbine Availability 8
- 4. Methodology 10
 - 4.1 Pipeline Architecture 10
 - 4.2 Feature Engineering 10
 - 4.3 Time Series Split & Validation 11
 - 4.4 Model Training 12
 - 4.5 Evaluation Metrics 13
- 5. Results 14
 - 5.1 Model Performance Overview 14
 - 5.2 Horizon Decay Analysis 15
 - 5.3 Model Comparison
 - 5.4 Farm-Level Results 17
 - 5.5 TB12 Turbine Analysis 19
 - 5.6 Operational Analysis 21
 - 5.7 Alert Accuracy 21
 - 5.8 Validation Charts 25
 - 5.9 Full Backtest Results 28
 - 5.10 Prediction Interval Calibration 29
 - 5.11 NWP Integration Status 30

6. API & Dashboard 31

6.1 System Architecture 31

6.2 API Endpoints 32

7. Output Files (Doc Section 15 Compliance) 33

8. Requirements Traceability Matrix 35

9. Source Code & Reproducible Configuration 37

9.1 Project Structure 37

9.2 Dependencies 38

9.3 End-to-End Reproduction 38

10. API Test Report 39

10.1 Endpoint Test Results 39

10.2 Schema Validation & Authentication 40

10.3 Latency & Resource Benchmark 40

11. Feature Status & Roadmap 41

11.1 Implemented Features 41

11.2 Prototype / Document-only 41

11.3 Planned Features 41

12. Conclusions & Future Work 42

12.1 Conclusions 42

12.2 Future Work 43

A. Appendix A. Response to Review Comments (v2.0.0) 43

B. Appendix B. Detailed Backtest Tables 47

1. Executive Summary

This report presents the design, implementation, and evaluation of a machine learning-based power forecasting system for the AMG Wind Farm. The farm consists of 12 turbines with a total installed capacity of 26.4 MW, located in Vietnam. The system provides multi-horizon forecasts at 10-minute, 30-minute, 1-hour, 6-hour, and 24-hour intervals using XGBoost and LightGBM models.

Key Results: The best-performing model achieves $R^2 = 0.9703$ (farm_total_power_target_10min (ridge, 10min)). The system includes **130 ML models** (130 = 13 targets x 2 algorithms x 5 horizons, counted from the actual *_model.joblib files), **10 baseline evaluations** (walk-forward persistence + ridge), and **520 model artifacts** (model + scaler + features + metadata per trained key). A FastAPI-based REST API serves 28 endpoints including real-time prediction, evaluation metrics, and alert generation. The system writes every output file according to the project output schema (Section 15: Dinh dang file dau ra) with defined column names, forecast_quality labels, and confidence-interval fields (calibration assessed in Section 5.10, not assumed at a nominal level).

Champion model per horizon — turbine-avg (min mean RMSE).

The champion is the model with the lowest mean RMSE within each horizon × level cell, computed directly from *evaluation_metrics.csv*; all four models compete (LightGBM, XGBoost, Ridge, persistence). The best ML model (LightGBM or XGBoost) per cell is listed for comparison.

Horizon	Level	Champion	RMSE (kW)	R2	Best ML	Best ML RMSE (kW)
10min	Turbine	Ridge	146.2	0.9419	LightGBM	154.2
30min	Turbine	Ridge	239.4	0.8442	LightGBM	246.3
1hour	Turbine	Ridge	306.1	0.7453	LightGBM	316.1
6hour	Turbine	LightGBM	453.7	0.4374	LightGBM	453.7
24hour	Turbine	Ridge	490.5	0.3443	LightGBM	511.2

Champion model per horizon — farm-total (min mean RMSE).

The champion is the model with the lowest mean RMSE within each horizon × level cell, computed directly from *evaluation_metrics.csv*; all four models compete (LightGBM, XGBoost, Ridge, persistence). The best ML model (LightGBM or XGBoost) per cell is listed for comparison.

Horizon	Level	Champion	RMSE (kW)	R2	Best ML	Best ML RMSE (kW)
10min	Farm	Ridge	1185.9	0.9703	XGBoost	1209.7
30min	Farm	Ridge	2303.2	0.8880	LightGBM	2347.7
1hour	Farm	Ridge	3117.4	0.7947	LightGBM	3198.7
6hour	Farm	LightGBM	4945.6	0.4797	LightGBM	4945.6
24hour	Farm	Ridge	5516.8	0.3525	LightGBM	5789.1

v2.1.0 revision note: This revision addresses the reviewer comments on report v2.0.0 (see Appendix A). Key changes: (1) data coverage is now reported on the observed raw timestamps (292,447 unique rows, 19,601 missing timestamps, 19,601 synthetic rows from 10-minute reindexing); (2) baseline training is

leakage-free and skill scores are computed on identical sample sets with explicit baselines; (3) a leakage audit, provenance inventory, sample trace, and farm-level bias analysis were added; (4) the API authenticates with an environment-variable API key (fail-closed) and CORS is restricted; (5) figure numbering is fixed.

Metric	Value
Total Turbines	12 (TB01 - TB12)
Total Capacity	26.4 MW
Raw Timestamps (unique)	292,447
Raw Missing Timestamps	19,601
Models Trained	130 ML models (13 targets x 2 algorithms x 5 horizons)
Baseline Evaluations	10 (walk-forward persistence + ridge)
Model Artifacts	520 (model + scaler + features + metadata per key)
Best R2	0.9703 (farm_total_power_target_10min (ridge, 10min))
Avg Availability	82.17%
Output Files	26 CSV files (Section 15 output schema)
API Endpoints	28 (FastAPI + Uvicorn)
API Tests	26 passing tests

2. Project Overview

2.1 Wind Farm Description

The AMG Wind Farm is a 26.4 MW wind power facility located in Vietnam, comprising 12 Vestas-class wind turbines (TB01 through TB12), each rated at 2,200 kW. Key turbine specifications:

Parameter	Value	Unit
Rated Power	2,200	kW
Cut-in Speed	3.0	m/s
Rated Speed	12.0	m/s
Cut-out Speed	25.0	m/s
Sampling Interval	10	minutes
Number of Turbines	12	-
Total Capacity	26,400	kW

2.2 Project Objectives

- Develop a multi-horizon power forecasting system for individual turbines and the entire farm
- Implement and compare XGBoost and LightGBM gradient boosting models
- Build a 15-step automated pipeline from raw SCADA data to forecast output
- Create a REST API with interactive dashboard for real-time forecasting
- Detect ramp events, anomalies, and turbine failure risks
- Produce output files following the project output schema (Section 15: Dinh dang file dau ra)
- Achieve $R^2 > 0.90$ at the 10-minute horizon for individual turbines

3. Data Description

3.1 SCADA Data Overview

The dataset consists of 11 semi-annual Excel files with overlapping coverage. Raw SCADA records span **2021-01-01 to 2026-12-07** at a nominal 10-minute sampling interval. After unioning all files and de-duplicating timestamps, the observed coverage is **292,447 unique timestamps** out of 312,048 expected steps over that span (coverage 93.72%; 19,601 missing timestamps; 0 duplicate rows removed). Each file contains 49 columns across 12 turbines.

Important data caveats (reported for transparency): (1) the raw source files extend beyond the report date (raw union end 2026-12-07), so the official evaluation window is explicitly cut at **evaluation_cutoff = min(report_date, raw_union_end) = 2026-08-01** (report date 2026-08-01). All test rows at/after the cutoff are flagged **is_simulated=1** in the processed data and are **excluded from the official evaluation** (18,576 rows set aside); every metric in this report is computed only on the official window (test window ends 2026-07-31). This is a supplier-side file mislabel: the last source file is named [01.2026-07.2026.xlsx](#) but its rows extend to 2026-12-07; the raw files are ingested as-is and the cutoff truncation is the pipeline's defense (flagged to the export owner, see README). (2) to obtain a regular 10-minute grid the pipeline re-indexes the timestamp axis, which introduces **19,601 synthetic rows** out of 312,048 processed timestamps (synthetic ratio 6.28%) — these rows are forward-filled with observed values and flagged **is_synthetic=1**; any row whose value was forward-filled is flagged **is_imputed=1**. Rows whose target is synthetic or imputed are **excluded from the official evaluation metrics** (Section 5.1) and from the leakage audit (Section 4.4).

Coverage Item		Value	
Processed timestamps (after reindex)		312,048	
Raw union timestamps		292,447	
Synthetic rows (reindexed, forward-filled)		19,601	
Synthetic ratio		6.28%	
Example synthetic timestamps		2026-01-08 07:00:00, 2026-01-08 07:10:00, 2026-01-08 07:20:00, 2026-01-08 07:30:00, 2026-01-08 07:40:00	
Sensor Group	Column Pattern	Unit	Valid Range
Wind Speed	TB{id}_Ambient WindSpeed Avg.	m/s	[0, 60]
Temperature	TB{id}_Ambient Temp. Avg.	deg C	[-10, 55]
Power (Target)	TB{id}_Grid Production Power Avg.	kW	[0, 2,200]
Frequency	TB{id}_Grid Production Frequency Avg.	Hz	[47, 53]

3.2 Data Quality Analysis

Data quality was assessed by computing missing rates per column. Most turbines show 6.4-7.1% missing data (classified as 'Minor gaps'), while TB05 has the highest turbine missing rate at 10.47% (classified as 'Moderate missing data'). The farm-level aggregate power column has 0.0% missing data; farm-total results below are therefore based on the observed aggregate series, and a missing value in the aggregate is not interpreted as a reliability statement.

Turbine	Missing Rate (%)	Status
TB01	6.49	Minor gaps
TB02	6.44	Minor gaps
TB03	6.87	Minor gaps
TB04	6.43	Minor gaps
TB05	10.47	Moderate missing data
TB06	6.40	Minor gaps
TB07	6.55	Minor gaps
TB08	6.41	Minor gaps
TB09	7.08	Minor gaps
TB10	6.61	Minor gaps
TB11	6.48	Minor gaps
TB12	6.92	Minor gaps

3.3 Turbine Availability

Availability is computed from the test period (test window ends 2026-07-31). To make the definition unambiguous, three complementary metrics are reported per turbine:

Metric	Formula	Meaning
Observed operational availability	$\text{Generating} / (\text{Generating} + \text{Stopped} + \text{Curtailed} + \text{Standby} + \text{Comm. loss}) \times 100$	Availability over the observed time only (missing telemetry excluded from the denominator).
Coverage-adjusted availability	$\text{Generating} / \text{Total elapsed hours} \times 100$	Conservative metric that treats missing time as unavailable (missing included in the denominator).
Data coverage	$\text{Observed hours} / \text{Total elapsed hours} \times 100$	Share of the calendar period with telemetry present (observed = generating + stopped + curtailed + standby + comm. loss).

Observed operational availability is the headline figure used throughout this report; coverage-adjusted availability and data coverage are reported alongside it so that a low headline value can be attributed to either low generation or missing telemetry (e.g. TB12, Section 5.5).

Availability was computed from the test period. The average availability across all turbines is 82.17%. TB01 has the highest at 84.25%, while TB12 has the lowest at 73.29%.

Turbine	Generating (hrs)	Stopped (hrs)	Missing (hrs)	Observed Operations Availability (%)	Adjusted Availability (%)	D(%) Cov. (%)
TB01	3,225	488	878	84.25	68.53	81.34
TB02	3,208	502	884	83.94	68.17	81.21
TB03	3,210	489	878	83.86	68.22	81.34
TB04	3,221	530	878	84.15	68.45	81.35
TB05	3,131	458	936	83.07	66.55	80.11
TB06	3,156	629	878	82.45	67.06	81.34
TB07	3,166	518	933	83.94	67.30	80.17
TB08	3,128	636	880	81.75	66.47	81.31
TB09	3,105	613	878	81.12	65.99	81.35
TB10	3,122	586	933	82.77	66.36	80.17
TB11	3,119	626	878	81.50	66.29	81.34
TB12	2,695	627	1,028	73.29	57.28	78.15

4. Methodology

4.1 Pipeline Architecture

The system implements a 15-step automated pipeline orchestrated by [main.py](#). Each step is modular, with dedicated source files in the [src/](#) directory. The pipeline handles data loading, preprocessing, feature engineering, model training, evaluation, and output generation.

Step	Module	Description
1	load_data.py	Load 11 raw SCADA Excel files
2	column_mapping.py	Standardize SCADA column names
3	data_validation.py	Validate ranges, units, completeness
3b	data_validation.py	Timestamp coverage audit, duplicates, timezone
4	preprocessing.py	Clean, resample, handle missing values
5	feature_engineering.py	Create lags, rolling stats, temporal features
6	split_time_series.py	Time-based 70/15/15 train/val/test split
7	train_baseline.py	Train persistence + linear regression baselines
8	train_power_model.py	Train XGBoost + LightGBM for all targets
9	train_anomaly_model.py	Isolation forest anomaly detection
10	train_failure_model.py	Failure risk analysis + availability
11	predict.py + evaluate.py	Generate predictions + compute metrics
12	predict.py	Create forecast output with confidence intervals
13	evaluate.py	Generate 6 summary visualizations
14	generate_outputs.py	Generate all doc Section 15 output files
15	inventory.py	Provenance — reindex additions + auto inventory

4.2 Feature Engineering

Feature engineering creates approximately 150 features per target variable from raw SCADA measurements. The feature categories include:

Category	Features	Count
Lag Features	1, 2, 3, 6, 12, 144 steps	6 per sensor
Rolling Statistics	Mean, Std, Min, Max over [6, 18, 36, 144]	16 per sensor
Temporal	Hour, Day of week, Month, Season, Is weekend	5
Power Change	Diff [1,3,6], Pct change [1]	4
Ramp Features	Rolling ramp rate, ramp indicators	4

Category	Features	Count
Interactions	Wind x temperature, power per wind	4+
Total per target	-	150 (ML) / 630 (Ridge)

4.3 Time Series Split & Validation

Data is split chronologically (70% train / 15% validation / 15% test) to prevent look-ahead bias. The split is strictly time-based; no random shuffle is applied. All feature engineering uses only data available up to the forecast issue time (shift()) for lags, rolling windows computed on past data only). Scaler, imputer, and feature selectors are fit exclusively on the training set.

Train / Validation / Test Split (row provenance):

Provenance per row is set by the pipeline (is_observed / is_synthetic / is_imputed, Section 3.1): “Observed” = timestamp present in the raw union; “Synthetic” = row inserted by the 10-min reindex; “Imputed” = row whose value was forward-filled. Only rows that are observed and not imputed are eligible to back the official metrics.

Split	Rows	Observed	Synthetic	Imputed	Obs. not imputed	Date Range
Train (70%)	218,433	218,433	0	2,395	216,038	2021-01-01 to 2025-02-25
Validation (15%)	46,807	46,087	720	679	45,420	2025-02-25 to 2026-01-16
Test (15%)	28,232	22,887	5,345	501	22,465	2026-01-16 to 2026-07-31

Valid targets per horizon (honest scoring denominators):

Split	Horizon	Rows	Valid Targets	Valid Ratio
Train	10min	218,433	217,879	99.75%
Train	30min	218,433	217,879	99.75%
Train	1hour	218,433	217,879	99.75%
Train	6hour	218,433	217,879	99.75%
Train	24hour	218,433	217,879	99.75%
Validation	10min	46,807	45,865	97.99%
Validation	30min	46,807	45,865	97.99%
Validation	1hour	46,807	45,865	97.99%
Validation	6hour	46,807	45,865	97.99%
Validation	24hour	46,807	45,865	97.99%
Test	10min	28,232	22,964	81.34%
Test	30min	28,232	22,964	81.34%
Test	1hour	28,232	22,964	81.34%
Test	6hour	28,232	22,964	81.34%
Test	24hour	28,232	22,964	81.34%

Data Leakage Prevention:

- Missing value imputation uses forward-fill (no future data in fill)
- Rolling statistics use only past observations (no center=True)
- StandardScaler fit on training data only, transforms on val/test
- Target shift correctly: $X(t) \rightarrow P(t+h)$; no power at t used as a feature for t
- Lag features use shift(1) so only past observations are consumed
- All splits are chronological; no random shuffle applied
- Feature selection (SelectKBest) fit on training data only

Walk-Forward Validation:

To assess model stability, walk-forward validation with 5 chronological folds is performed on baseline models (Persistence and Ridge regression). Each fold uses an expanding training window with a held-out test segment. Mean and standard deviation of RMSE and R^2 across folds are reported to quantify performance variability. The 5 folds are the count actually produced by the expanding-window split (walk_forward_split), not a requested target.

4.4 Model Training

Two gradient boosting algorithms were used: XGBoost and LightGBM. Both models were trained with identical hyperparameters for fair comparison. Each target variable (12 turbines + farm aggregate x 5 horizons) received independent model training, resulting in **130 ML models** (13 targets x 2 algorithms x 5 horizons). This count is read from the actual *_model.joblib files in models/ via inventory_summary.json, not computed by multiplication. Each trained key is stored as 4 artifacts (model + scaler + feature list + metadata), i.e. **520 files**; the 10 baseline evaluations (persistence + ridge x 5 horizons) are trained and evaluated separately via walk-forward validation.

Hyperparameter	XGBoost	LightGBM
n_estimators	150	150
max_depth	6	6
learning_rate	0.1	0.1
subsample	0.8	0.8
colsample_bytree	0.5	0.5
random_state	42	42

Hyperparameter tuning (evidence trail): the pipeline exposes an optional tuning stage ([training.tuning.enabled](#) in config.yaml, default **false**). When enabled, it runs a 12-iteration Bayesian-style random search on XGBoost and LightGBM using TimeSeriesSplit cross-validation (3 folds, expanding window) per turbine-horizon target, records the per-fold and mean CV RMSE for every trial, and persists two evidence files: [data/metadata/tuning_results.csv](#) (per-trial, per-fold, mean CV RMSE) and [data/metadata/best_params.json](#) (winning hyperparameters per target). The shipped models in this report use the fixed hyperparameters in the table above (tuning disabled); re-running [main.py](#) with tuning enabled replaces those parameters and records the full evidence trail for re-audit.

Leakage Audit (every trained model):

The pipeline runs an automated leakage audit over every trained model. Each model's persisted feature list is checked for any target/future marker (e.g. '_target_', '_missing_', '_status_', '_is_anomaly_', '_anomaly_score_', '_is_stopped_', '_failure_event_'); the audit also verifies that the target column $P(t+h)$ is not among the features, that no future-leaning feature is present, that timestamps satisfy $\text{timestamp_target} = \text{timestamp_issue} + \text{horizon}$, and that no two models return predictions identical to their target. The audit covers **180 model fits** across ridge, XGBoost and LightGBM (120 ML fits), each checked per turbine and per horizon (10 min/30 min/1 h/6 h/24 h). Breakdown: ridge: 60 models, 0 flagged; xgboost: 60 models, 0 flagged; lightgbm: 60 models, 0 flagged. Result (full audit, every turbine x horizon x model family): **0 flagged models out of 180** audited (55,800 features used in total). A non-zero count aborts the pipeline (fail-closed); the per-family audit (leakage_audit_full.csv) is persisted to [data/metadata/leakage_audit_full.csv](#).

4.5 Evaluation Metrics

Metric	Formula	Description
MAE	$\text{Mean}(F - O)$	Mean Absolute Error (kW)
nMAE	$\text{MAE} / P_{\text{rated}} \times 100\%$	Normalized MAE (%)
RMSE	$\sqrt{\text{Mean}((F - O)^2)}$	Root Mean Square Error (kW)
nRMSE	$\text{RMSE} / P_{\text{rated}} \times 100\%$	Normalized RMSE (%)
Bias	$\text{Mean}(F - O)$	Mean Error (kW), detects over/under-forecast
R2	$1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$	Coefficient of Determination
Skill Score	$1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{baseline}}$	Improvement over a stated baseline (0 = same, 1 = perfect)
n_samples	$\text{count}(\text{valid } P(t+h))$	Explicit scoring denominator per horizon
Max Error	$\max(F - O)$	Worst-case error (kW)

Per P0-03 review comment, every skill score is reported with its **explicit baseline** and its **sample size**: 'Skill vs persistence' = $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{persistence}}$ and 'Skill vs Ridge' = $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{ridge}}$, both computed on the **identical set of test samples** where all three models (boosting, persistence, Ridge) have a valid target $P(t+h)$. n_samples is stored per row in evaluation_metrics.csv.

5. Results

5.1 Model Performance Overview

The following table summarizes mean turbine R² (average of 12 individual turbine R² values, from *metrics.csv*) per model per horizon. Ridge is the champion (min mean RMSE) in 4 of 5 turbine cells. LightGBM is the champion in 1 of 5 turbine cells (6hour). At turbine level, LightGBM and XGBoost mean R² differ by at most 0.014 across horizons. Best mean turbine R² degrades from 0.942 at 10-min to 0.344 at 24-hour, consistent with the inherent difficulty of longer-range wind power prediction.

Horizon	Model	MAE	nMAE%	RMSE	nRMSE%	Bias	R2	SkIP	SkIR	n
10min	lightgbm	98.2	4.5	154.2	7.0	+5.0	0.9338	0.012	-0.055	22,306
10min	persistence	92.0	4.2	156.2	7.1	+0.2	0.9336	0.000	-0.069	22,304
10min	ridge	91.5	4.2	146.2	6.6	-1.2	0.9419	0.064	0.000	22,306
10min	xgboost	99.7	4.5	155.3	7.1	+6.0	0.9330	0.005	-0.062	22,306
1hour	lightgbm	224.1	10.2	316.1	14.4	+4.5	0.7278	0.082	-0.033	22,311
1hour	persistence	213.6	9.7	344.5	15.7	+1.2	0.6778	0.000	-0.126	22,304
1hour	ridge	211.6	9.6	306.1	13.9	-3.4	0.7453	0.111	0.000	22,311
1hour	xgboost	230.8	10.5	320.0	14.5	+12.7	0.7208	0.071	-0.046	22,311
24hour	lightgbm	405.1	18.4	511.2	23.2	+15.5	0.2883	0.129	-0.043	21,415
24hour	persistence	396.2	18.0	585.5	26.6	-7.7	0.0650	0.000	-0.198	21,315
24hour	ridge	378.6	17.2	490.5	22.3	-6.0	0.3443	0.164	0.000	21,415
24hour	xgboost	412.8	18.8	516.2	23.5	+29.0	0.2742	0.120	-0.054	21,415
30min	lightgbm	167.3	7.6	246.3	11.2	+8.1	0.8344	0.055	-0.029	22,308
30min	persistence	156.8	7.1	260.6	11.8	+0.6	0.8155	0.000	-0.089	22,304
30min	ridge	158.0	7.2	239.4	10.9	-2.3	0.8442	0.081	0.000	22,308
30min	xgboost	173.1	7.9	250.5	11.4	+13.6	0.8284	0.038	-0.047	22,308
6hour	lightgbm	349.4	15.9	453.7	20.6	+10.8	0.4374	0.371	0.023	22,159
6hour	persistence	518.5	23.6	723.5	32.9	-1.6	-0.4257	0.000	-0.560	22,124
6hour	ridge	358.8	16.3	464.9	21.1	+9.3	0.4094	0.356	0.000	22,159
6hour	xgboost	355.6	16.2	457.8	20.8	+19.1	0.4276	0.366	0.015	22,159

Mean of the 12 turbine rows per (horizon, model) cell of *metrics.csv*. SkIP = skill vs persistence; SkIR = skill vs Ridge; both on identical samples (Section 4.5). n = mean valid test samples per row.

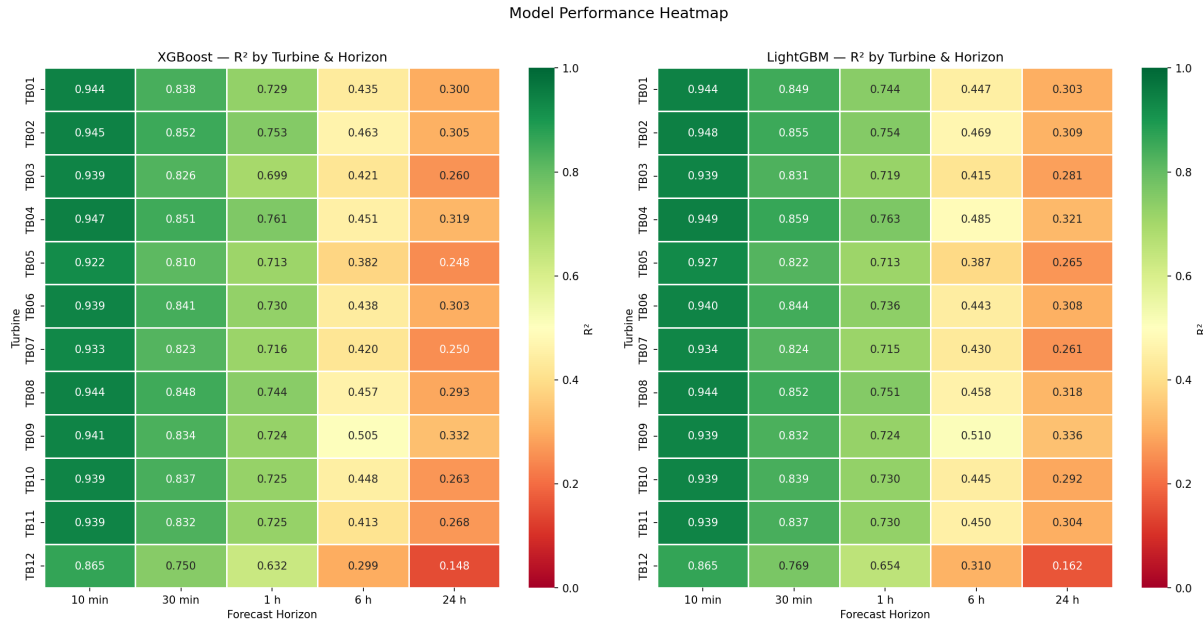


Figure 1: Model performance heatmap across turbines and horizons

Walk-Forward Validation (Baselines):

Walk-forward validation across 5 chronological folds (baselines). Values are mean +/- standard deviation across folds.

Model	Horizon	RMSE (kW)	R2
Persistence	10min	267.4 +/- 333.4	0.9253 +/- 0.0134
Persistence	30min	472.9 +/- 651.7	0.7904 +/- 0.0276
Persistence	1hour	629.1 +/- 902.5	0.6466 +/- 0.0370
Persistence	6hour	1224.3 +/- 1871.0	-0.2346 +/- 0.1078
Persistence	24hour	1187.8 +/- 1811.2	-0.1642 +/- 0.0883
Ridge	10min	263.3 +/- 318.0	0.9223 +/- 0.0473
Ridge	30min	449.4 +/- 620.8	0.8108 +/- 0.0331
Ridge	1hour	583.2 +/- 833.0	0.6921 +/- 0.0774
Ridge	6hour	914.9 +/- 1387.6	0.2988 +/- 0.1280
Ridge	24hour	1023.8 +/- 1577.7	0.1432 +/- 0.1094

Key evidence: Persistence 10min: RMSE=267.4+/-333.4 kW, R2=0.9253+/-0.0134; Persistence 1hour: RMSE=629.1+/-902.5 kW, R2=0.6466+/-0.0370; Persistence 24hour: RMSE=1187.8+/-1811.2 kW, R2=-0.1642+/-0.0883; Persistence 30min: RMSE=472.9+/-651.7 kW, R2=0.7904+/-0.0276.

5.2 Horizon Decay Analysis

As expected, forecast accuracy decreases with increasing horizon. The R² decay is approximately linear from 10-min to 1-hour, then accelerates for 6-hour and 24-hour horizons. This is consistent with the inherent difficulty of longer-range wind power prediction.

Evidence: 10min lightgbm: $R^2=0.9338\pm0.0226$; 10min persistence: $R^2=0.9336\pm0.0054$; 10min ridge: $R^2=0.9419\pm0.0052$; 10min xgboost: $R^2=0.9330\pm0.0223$; 1hour lightgbm: $R^2=0.7278\pm0.0281$; 1hour persistence: $R^2=0.6778\pm0.0178$.

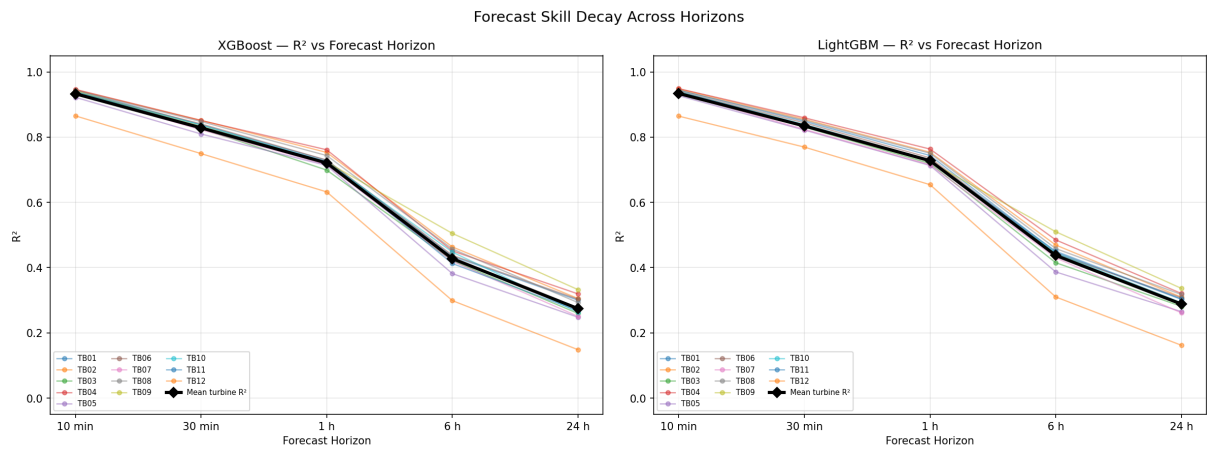


Figure 2: R2 degradation across forecast horizons

5.3 Model Comparison (turbine-avg)

XGBoost vs LightGBM mean turbine R^2 gap per horizon: 10min: $|\Delta R^2|=0.0008$; 1hour: $|\Delta R^2|=0.0070$; 24hour: $|\Delta R^2|=0.0141$; 30min: $|\Delta R^2|=0.0060$; 6hour: $|\Delta R^2|=0.0098$.

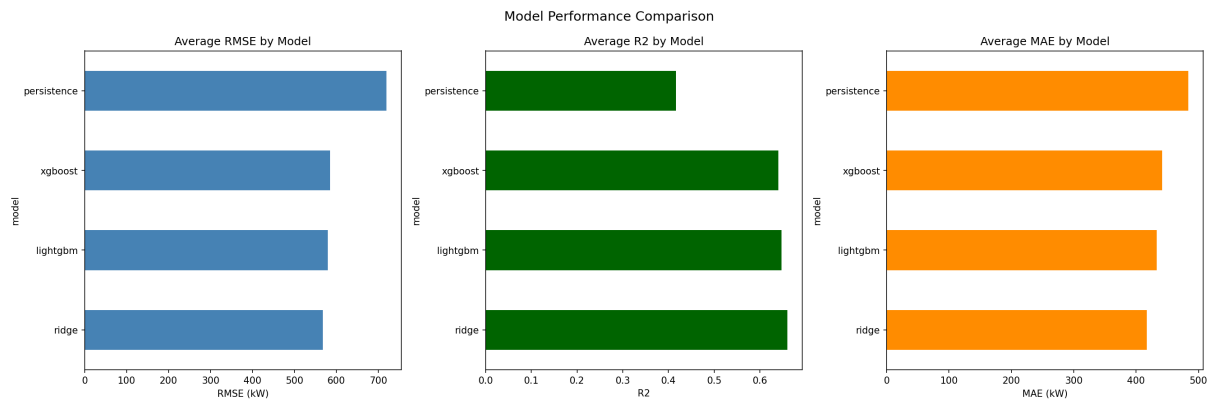


Figure 3: XGBoost vs LightGBM comparison by horizon

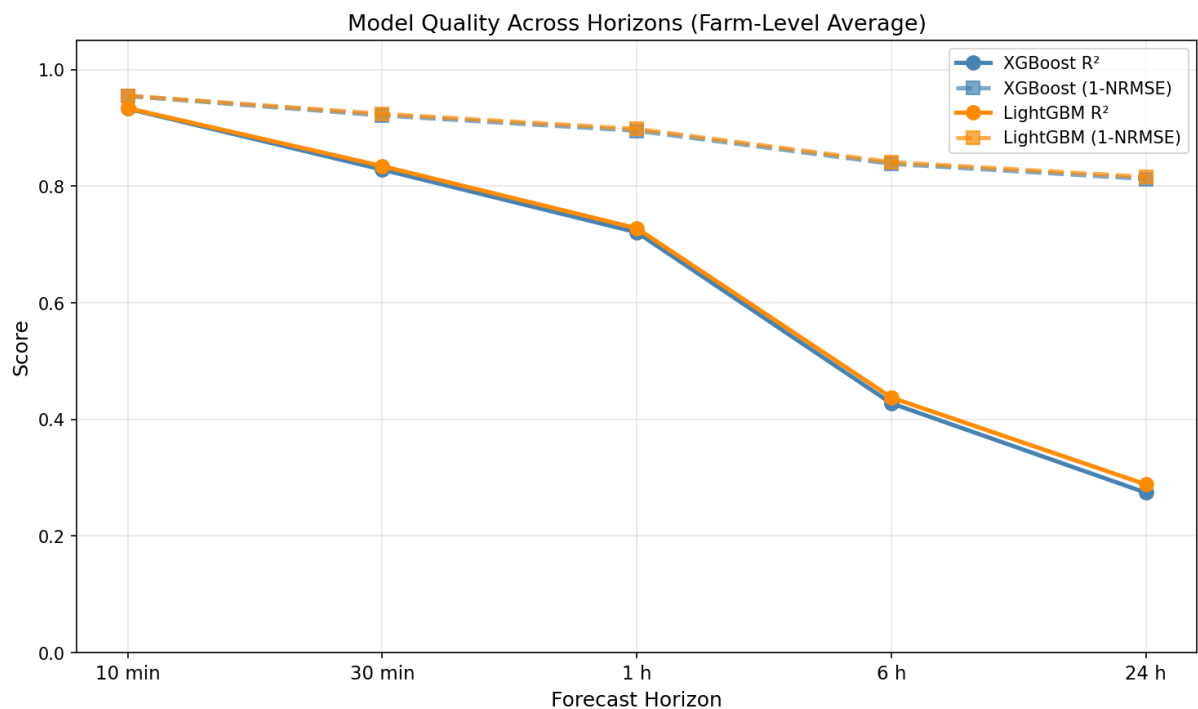


Figure 4: Multi-metric radar comparison of model performance

5.4 Farm-Level Results

Farm-level metrics are calculated **directly** on the summed farm total power (actual and predicted), not as an average of individual turbine R^2 . This gives a proper assessment of farm-level forecast accuracy.

Horizon	Model	MAE (kW)	RMSE (kW)	nRMSE%	Bias	R2	R2 corr.	Bias corr.
10min	xgboost	800.6	1209.7	4.6	+106.1	0.9691	0.9693	+6.7
10min	lightgbm	794.0	1216.2	4.6	+104.7	0.9688	0.9688	+39.1

Horizon	Model	MAE (kW)	RMSE (kW)	nRMSE%	Bias	R2	R2 corr.	Bias corr.
30min	xgboost	1728.3	2453.9	9.3	+268.0	0.8728	0.8748	-83.9
30min	lightgbm	1591.0	2347.7	8.9	+130.5	0.8836	0.8840	+45.1
1hour	xgboost	2342.0	3231.1	12.2	+186.0	0.7794	0.7828	-18.2
1hour	lightgbm	2257.7	3198.7	12.1	+77.0	0.7838	0.7856	+87.5
6hour	xgboost	3918.7	4962.3	18.8	+350.5	0.4762	0.4836	-129.4
6hour	lightgbm	3873.4	4945.6	18.7	+198.4	0.4797	0.4843	-63.5
24hour	xgboost	4669.0	5791.9	21.9	+560.2	0.2863	0.2821	+237.2
24hour	lightgbm	4682.8	5789.1	21.9	+545.7	0.2870	0.2828	+179.5

Evidence: Farm-level R^2 ranges from 0.9691 (10min). The 10-min farm forecast achieves $R^2=0.9691$ with MAE=801 kW (nMAE=3.0%).

Farm-total champions per horizon (min mean RMSE from *farm_metrics.csv*): 10min: XGBoost RMSE 1210 kW ($R^2=0.9691$); 1hour: LightGBM RMSE 3199 kW ($R^2=0.7838$); 24hour: LightGBM RMSE 5789 kW ($R^2=0.2870$); 30min: LightGBM RMSE 2348 kW ($R^2=0.8836$); 6hour: LightGBM RMSE 4946 kW ($R^2=0.4797$).

At farm-total level, LightGBM and XGBoost R^2 differ by at most 0.011 across horizons (from *farm_metrics.csv*).

Bias correction (P1-04): a per-model linear offset fitted on the validation split (corrected = slope $\times$ predicted + intercept) was applied to the test forecasts. Best corrected $R^2=0.9693$ (10min, xgboost); raw/corrected columns are shown side by side above and every corrected metric is scored on the same samples as the raw one.

5.4.1 Farm Bias Analysis (P1-04)

Model bias is examined by comparing the direct farm-total forecast with the SUM of the 12 individual turbine forecasts, and against the observed farm power ($P(t+h)$). A positive bias means the farm model over-forecasts on average. This directly addresses the review concern that reported skill may hide systematic under-/over-forecasting.

Horizon	n	Actual (kW)	Farm model (kW)	Bias (kW)	Bias % rated	MAE (kW)	Farm vs sum (kW)
10min	28,232	5,536	5,783	+247.0	+0.936	813.8	-59.5
30min	28,232	5,537	6,644	+1,107.0	+4.193	2,299.4	+464.0
1hour	28,232	5,538	6,229	+690.8	+2.617	2,459.6	+61.5
6hour	28,232	5,549	6,998	+1,448.7	+5.488	4,393.7	-102.9
24hour	28,232	5,598	6,569	+970.7	+3.677	4,670.3	-504.3

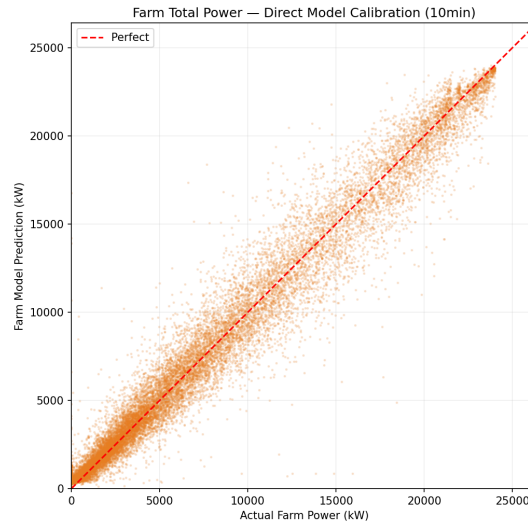


Figure 5: Farm-level bias vs predicted power with calibration reference (identity line)

5.4.2 Horizon Comparison Window Check (P1-04)

To rule out a sampling artifact, R^2 for every horizon pair is recomputed on the **intersection** of valid target samples — i.e. the same test window (same timestamps) and the same `n_at_capacity` / `n_zero_power` masks. If the 24h > 6h R^2 result survives on identical samples, it is a genuine forecast-quality effect rather than a target-shift artifact.

Horizon A	Horizon B	n common	Window same	R2 A (common)	R2 B (common)	Delta R2
10min	24hour	28,232	yes	0.9702	0.2848	-0.6853
30min	24hour	28,232	yes	0.7966	0.2848	-0.5117
1hour	24hour	28,232	yes	0.7752	0.2848	-0.4904
6hour	24hour	28,232	yes	0.3656	0.2848	-0.0808

5.5 TB12 Turbine Analysis

Turbine TB12 shows significantly lower forecast accuracy compared to other turbines. In the official test window (28,232 rows) it has 21.85% missing data, 20.87% stopped/near-zero power output, and a 19.84% frozen-data ratio (225 blocks). Across the full processed dataset TB12_power is 6.92% missing after forward-fill (per-column, Section 3.2); the test-window rate reflects the evaluation period only. Its wind-power correlation within the official window is 0.8402. These data-quality issues warrant investigation into sensor calibration and operating conditions.

Metric	Value
Missing Data Rate (%)	21.85
Stopped / Near-Zero Power (%)	20.87
Frozen/Stuck Data Ratio (%)	19.84
Frozen Data Blocks	225
Mean Power (kW)	461.7

Metric	Value
Std Power (kW, when operating)	564.6
Mean Power TB09 (kW)	553.8
Mean Power TB04 (kW)	620.4
Power Correlation with TB09	0.7647
Power Correlation with TB04	0.7905
Mean Wind Speed TB09 (m/s)	5.84
Wind-Power Correlation TB12	0.8402
R2 10min (lightgbm)	0.8645
R2 10min (persistence)	0.9267
R2 10min (ridge)	0.931
R2 10min (xgboost)	0.8653
R2 1hour (lightgbm)	0.6543
R2 1hour (persistence)	0.6643
R2 1hour (ridge)	0.7219
R2 1hour (xgboost)	0.6321
R2 24hour (lightgbm)	0.1619
R2 24hour (persistence)	-0.1313
R2 24hour (ridge)	0.2193
R2 24hour (xgboost)	0.1483
R2 30min (lightgbm)	0.7695
R2 30min (persistence)	0.8021
R2 30min (ridge)	0.8247
R2 30min (xgboost)	0.7496
R2 6hour (lightgbm)	0.3101
R2 6hour (persistence)	-0.4069
R2 6hour (ridge)	0.3043
R2 6hour (xgboost)	0.2988

TB12 shows higher missing data rate and longer stop periods compared to TB09 and TB04. Its power-wind correlation is weaker, suggesting possible sensor drift or wake effects. A dedicated sensor recalibration and data quality investigation is recommended.

5.6 Operational Analysis

Error analysis by operating regime provides insight into model behavior under different conditions. The model comparison and radar summary above show consistent performance characteristics across turbines.

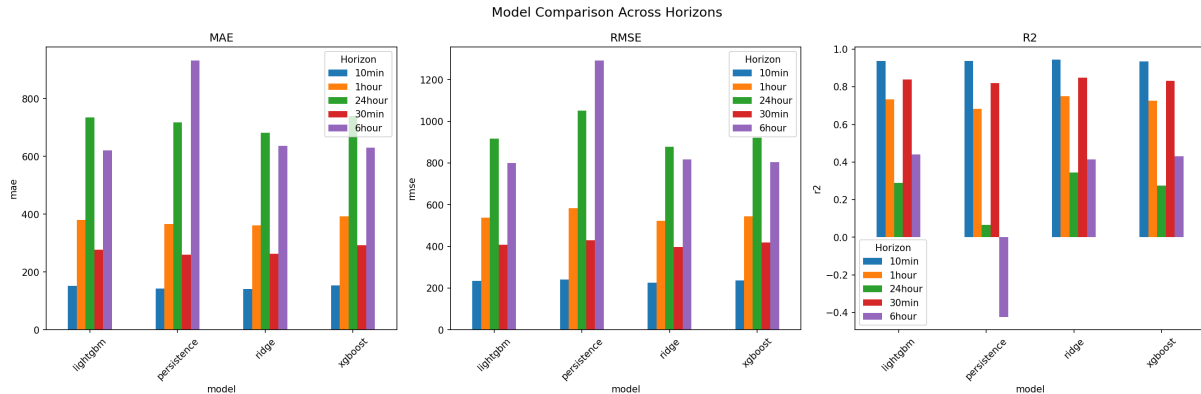


Figure 6: Horizon-wise model performance comparison

5.7 Alert Accuracy

Ramp event detection accuracy is evaluated using Precision, Recall, F1-score, and False Alarm Rate (FAR). A ramp event is defined as a power change exceeding 0.5% per minute of rated capacity.

Alert semantics: all ramp, anomaly, temperature and failure-risk alerts are **informational advisories** produced by heuristic screening (`method=heuristic_screening`, `confirmed=False`, `verification_status=SCREENING_ONLY` in every alert row and in `data/metadata/alert_screening_summary.json`). They do not replace operator decisions, SCADA trip logic, or safety interlocks, and they never modify generation set-points. Accuracy metrics in this section therefore measure the detection quality of the advisory stream against the heuristic ground-truth rules, not an operational control function or confirmed event forecasts.

Evidence: Average Precision=0.805, Recall=0.855, F1=0.829 across all turbine-horizon-model combinations.

Model	Precision	Recall	F1	FAR	TP	FP	FN
xgboost_TB01_10min	0.883	0.910	0.897	0.117	15569	2053	1530
lightgbm_TB01_10min	0.897	0.915	0.906	0.102	15654	1788	1445
xgboost_TB01_30min	0.816	0.890	0.851	0.184	15217	3443	1882
lightgbm_TB01_30min	0.850	0.892	0.871	0.150	15259	2696	1840
xgboost_TB01_1hour	0.812	0.883	0.846	0.188	15091	3499	2008
lightgbm_TB01_1hour	0.838	0.878	0.857	0.162	15009	2901	2090
xgboost_TB01_6hour	0.744	0.814	0.777	0.256	13924	4793	3175
lightgbm_TB01_6hour	0.770	0.802	0.786	0.230	13720	4090	3379
xgboost_TB01_24hour	0.756	0.829	0.791	0.244	14176	4579	2923
lightgbm_TB01_24hour	0.772	0.801	0.786	0.228	13698	4052	3401

Model	Precision	Recall	F1	FAR	TP	FP	FN
xgboost_TB02_10min	0.872	0.910	0.890	0.128	15446	2269	1535
lightgbm_TB02_10min	0.900	0.909	0.905	0.100	15440	1715	1541
xgboost_TB02_30min	0.854	0.878	0.866	0.146	14904	2547	2077
lightgbm_TB02_30min	0.858	0.877	0.868	0.142	14897	2458	2084
xgboost_TB02_1hour	0.846	0.871	0.858	0.154	14791	2703	2190
lightgbm_TB02_1hour	0.850	0.870	0.860	0.150	14773	2617	2208
xgboost_TB02_6hour	0.745	0.816	0.779	0.255	13851	4732	3130
lightgbm_TB02_6hour	0.769	0.789	0.779	0.231	13399	4031	3582
xgboost_TB02_24hour	0.754	0.830	0.790	0.245	14085	4582	2896
lightgbm_TB02_24hour	0.770	0.795	0.782	0.230	13507	4035	3474
xgboost_TB03_10min	0.877	0.921	0.898	0.123	15821	2221	1363
lightgbm_TB03_10min	0.894	0.915	0.905	0.106	15727	1859	1457
xgboost_TB03_30min	0.842	0.888	0.864	0.158	15258	2867	1926
lightgbm_TB03_30min	0.875	0.889	0.882	0.125	15275	2179	1909
xgboost_TB03_1hour	0.796	0.882	0.837	0.204	15162	3876	2022
lightgbm_TB03_1hour	0.829	0.878	0.853	0.171	15087	3111	2097
xgboost_TB03_6hour	0.767	0.806	0.786	0.233	13849	4195	3335
lightgbm_TB03_6hour	0.782	0.783	0.782	0.218	13450	3754	3734
xgboost_TB03_24hour	0.766	0.833	0.798	0.234	14318	4383	2866
lightgbm_TB03_24hour	0.778	0.803	0.790	0.222	13805	3943	3379
xgboost_TB04_10min	0.882	0.914	0.898	0.118	15613	2084	1469
lightgbm_TB04_10min	0.902	0.921	0.911	0.098	15725	1710	1357
xgboost_TB04_30min	0.837	0.888	0.862	0.163	15175	2951	1907
lightgbm_TB04_30min	0.848	0.889	0.868	0.152	15185	2716	1897
xgboost_TB04_1hour	0.818	0.879	0.848	0.182	15020	3337	2062
lightgbm_TB04_1hour	0.837	0.878	0.857	0.163	15000	2912	2082
xgboost_TB04_6hour	0.753	0.827	0.788	0.247	14126	4634	2956
lightgbm_TB04_6hour	0.778	0.796	0.787	0.222	13592	3868	3490
xgboost_TB04_24hour	0.762	0.850	0.803	0.238	14518	4545	2564
lightgbm_TB04_24hour	0.773	0.815	0.794	0.227	13928	4083	3154
xgboost_TB05_10min	0.841	0.928	0.882	0.160	15306	2904	1188
lightgbm_TB05_10min	0.860	0.923	0.890	0.140	15232	2483	1262
xgboost_TB05_30min	0.793	0.900	0.844	0.206	14846	3863	1648

Model	Precision	Recall	F1	FAR	TP	FP	FN
lightgbm_TB05_30min	0.826	0.901	0.862	0.174	14859	3142	1635
xgboost_TB05_1hour	0.807	0.887	0.845	0.193	14628	3498	1866
lightgbm_TB05_1hour	0.805	0.888	0.845	0.195	14651	3539	1843
xgboost_TB05_6hour	0.722	0.820	0.768	0.278	13521	5203	2973
lightgbm_TB05_6hour	0.741	0.812	0.775	0.260	13393	4694	3101
xgboost_TB05_24hour	0.733	0.838	0.782	0.267	13827	5037	2667
lightgbm_TB05_24hour	0.745	0.791	0.767	0.255	13047	4456	3447
xgboost_TB06_10min	0.900	0.907	0.903	0.100	15340	1703	1572
lightgbm_TB06_10min	0.903	0.911	0.907	0.097	15406	1652	1506
xgboost_TB06_30min	0.847	0.870	0.859	0.153	14715	2652	2197
lightgbm_TB06_30min	0.848	0.870	0.859	0.152	14713	2640	2199
xgboost_TB06_1hour	0.806	0.859	0.832	0.194	14524	3500	2388
lightgbm_TB06_1hour	0.841	0.859	0.850	0.159	14523	2742	2389
xgboost_TB06_6hour	0.755	0.806	0.780	0.245	13635	4426	3277
lightgbm_TB06_6hour	0.770	0.771	0.771	0.230	13047	3886	3865
xgboost_TB06_24hour	0.758	0.834	0.794	0.242	14096	4494	2816
lightgbm_TB06_24hour	0.780	0.789	0.784	0.220	13339	3765	3573
xgboost_TB07_10min	0.882	0.908	0.895	0.118	15320	2054	1548
lightgbm_TB07_10min	0.895	0.916	0.905	0.105	15453	1816	1415
xgboost_TB07_30min	0.834	0.876	0.855	0.166	14782	2947	2086
lightgbm_TB07_30min	0.856	0.881	0.868	0.144	14855	2507	2013
xgboost_TB07_1hour	0.840	0.862	0.851	0.160	14539	2764	2329
lightgbm_TB07_1hour	0.844	0.866	0.855	0.156	14599	2696	2269
xgboost_TB07_6hour	0.746	0.808	0.776	0.254	13623	4628	3245
lightgbm_TB07_6hour	0.764	0.772	0.768	0.236	13018	4018	3850
xgboost_TB07_24hour	0.750	0.830	0.788	0.250	14006	4679	2862
lightgbm_TB07_24hour	0.765	0.801	0.783	0.235	13515	4144	3353
xgboost_TB08_10min	0.901	0.903	0.902	0.099	15094	1664	1621
lightgbm_TB08_10min	0.901	0.905	0.903	0.099	15133	1653	1582
xgboost_TB08_30min	0.845	0.872	0.858	0.155	14571	2664	2144
lightgbm_TB08_30min	0.851	0.876	0.864	0.149	14651	2560	2064
xgboost_TB08_1hour	0.804	0.874	0.838	0.196	14614	3565	2101
lightgbm_TB08_1hour	0.856	0.867	0.862	0.144	14496	2438	2219

Model	Precision	Recall	F1	FAR	TP	FP	FN
xgboost_TB08_6hour	0.739	0.801	0.769	0.261	13390	4726	3325
lightgbm_TB08_6hour	0.769	0.769	0.769	0.231	12855	3862	3860
xgboost_TB08_24hour	0.754	0.835	0.792	0.246	13957	4563	2758
lightgbm_TB08_24hour	0.768	0.806	0.787	0.232	13481	4076	3234
xgboost_TB09_10min	0.869	0.911	0.889	0.131	14993	2260	1465
lightgbm_TB09_10min	0.891	0.909	0.900	0.109	14965	1839	1493
xgboost_TB09_30min	0.832	0.884	0.857	0.168	14551	2945	1907
lightgbm_TB09_30min	0.844	0.882	0.863	0.156	14524	2680	1934
xgboost_TB09_1hour	0.825	0.876	0.850	0.175	14414	3063	2044
lightgbm_TB09_1hour	0.832	0.873	0.852	0.168	14374	2899	2084
xgboost_TB09_6hour	0.728	0.773	0.750	0.272	12724	4764	3734
lightgbm_TB09_6hour	0.748	0.757	0.753	0.252	12461	4195	3997
xgboost_TB09_24hour	0.739	0.845	0.788	0.262	13912	4926	2546
lightgbm_TB09_24hour	0.753	0.820	0.785	0.247	13499	4424	2959
xgboost_TB10_10min	0.895	0.912	0.904	0.104	15435	1802	1485
lightgbm_TB10_10min	0.902	0.908	0.905	0.098	15367	1664	1553
xgboost_TB10_30min	0.855	0.884	0.869	0.145	14950	2540	1970
lightgbm_TB10_30min	0.855	0.880	0.867	0.145	14891	2523	2029
xgboost_TB10_1hour	0.806	0.879	0.841	0.194	14881	3587	2039
lightgbm_TB10_1hour	0.879	0.871	0.875	0.121	14739	2028	2181
xgboost_TB10_6hour	0.750	0.791	0.770	0.250	13377	4460	3543
lightgbm_TB10_6hour	0.777	0.754	0.765	0.223	12757	3658	4163
xgboost_TB10_24hour	0.750	0.826	0.786	0.249	13968	4644	2952
lightgbm_TB10_24hour	0.776	0.796	0.786	0.224	13469	3889	3451
xgboost_TB11_10min	0.901	0.915	0.908	0.099	15293	1675	1423
lightgbm_TB11_10min	0.909	0.911	0.910	0.091	15231	1533	1485
xgboost_TB11_30min	0.832	0.884	0.857	0.169	14779	2994	1937
lightgbm_TB11_30min	0.839	0.883	0.861	0.161	14766	2825	1950
xgboost_TB11_1hour	0.810	0.875	0.841	0.190	14633	3439	2083
lightgbm_TB11_1hour	0.854	0.875	0.865	0.146	14631	2496	2085
xgboost_TB11_6hour	0.730	0.800	0.763	0.270	13370	4954	3346
lightgbm_TB11_6hour	0.755	0.759	0.757	0.245	12694	4121	4022
xgboost_TB11_24hour	0.738	0.839	0.785	0.262	14023	4967	2693

Model	Precision	Recall	F1	FAR	TP	FP	FN
lightgbm_TB11_24hour	0.754	0.803	0.778	0.246	13426	4389	3290
xgboost_TB12_10min	0.780	0.909	0.840	0.220	13152	3712	1314
lightgbm_TB12_10min	0.780	0.913	0.842	0.220	13215	3716	1251
xgboost_TB12_30min	0.710	0.876	0.784	0.290	12668	5179	1798
lightgbm_TB12_30min	0.727	0.874	0.793	0.273	12637	4756	1829
xgboost_TB12_1hour	0.687	0.864	0.765	0.313	12497	5687	1969
lightgbm_TB12_1hour	0.731	0.857	0.789	0.269	12396	4556	2070
xgboost_TB12_6hour	0.643	0.790	0.709	0.357	11431	6359	3035
lightgbm_TB12_6hour	0.652	0.756	0.700	0.348	10944	5842	3522
xgboost_TB12_24hour	0.645	0.831	0.726	0.355	12014	6611	2452
lightgbm_TB12_24hour	0.653	0.785	0.713	0.346	11357	6023	3109

5.8 Validation Charts

This section presents comprehensive model validation charts: predicted vs actual scatter, error distribution, residual analysis, and error breakdown by operating regime (wind speed, power region, season, and day/night).

5.8.1 Predicted vs Actual Scatter

Scatter plots of predicted vs actual power for the best-performing model at each horizon. Points close to the diagonal (red dashed line) indicate accurate predictions.

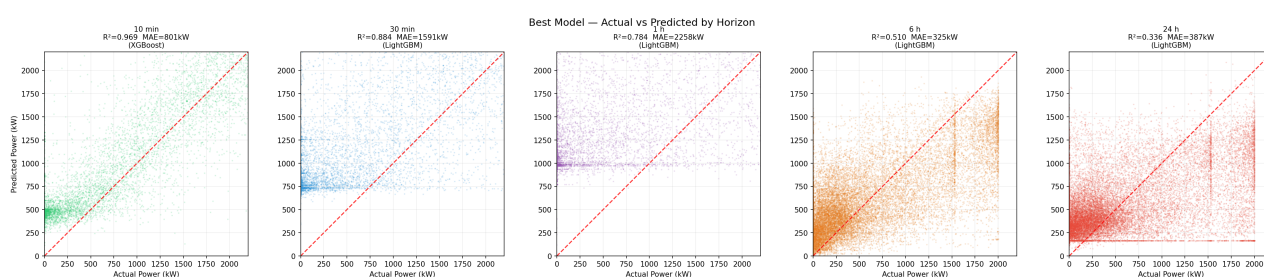


Figure 7: Best model predicted vs actual scatter by horizon

5.8.2 Error Distribution

Histogram of prediction errors (actual - predicted) for the best model at each horizon. Symmetric distribution centered near zero indicates unbiased forecasts.

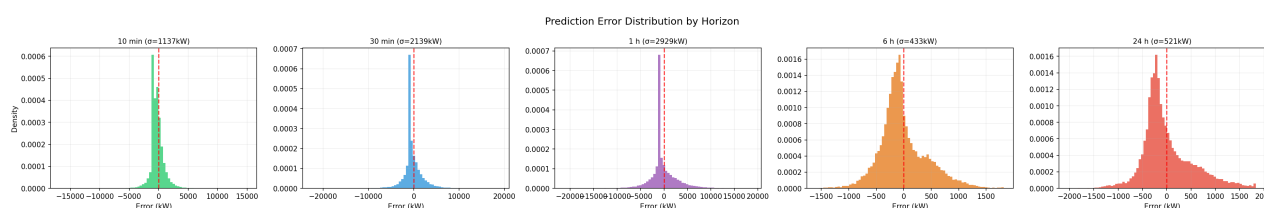


Figure 8: Prediction error distribution by horizon

5.8.3 Residual Analysis

Top row: residual vs predicted scatter — random scatter around zero indicates homoscedasticity. Bottom row: residual density histogram.

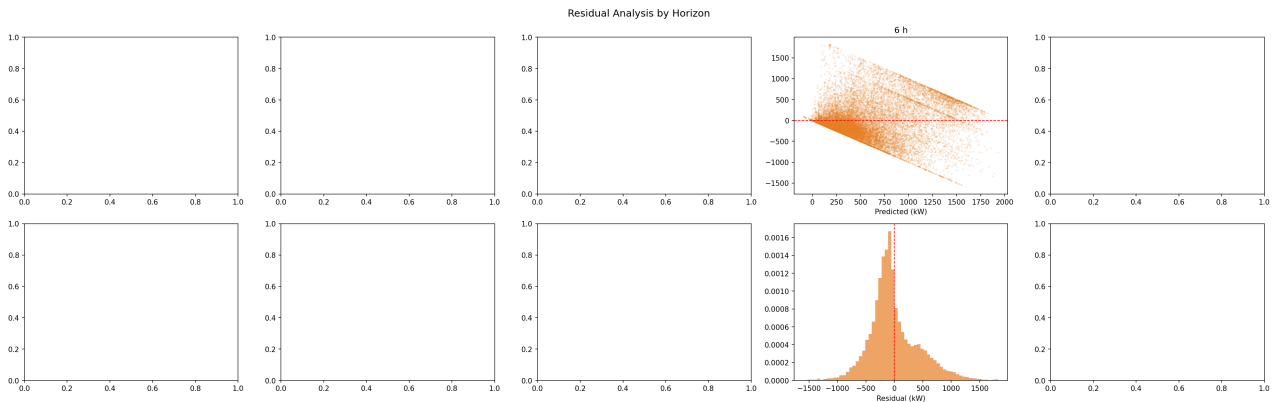


Figure 9: Residual analysis — scatter (top) and density (bottom)

5.8.4 Error by Operating Regime

Error analysis by wind speed bins, power output regions, seasons, and day vs night. This reveals systematic biases under specific operating conditions.

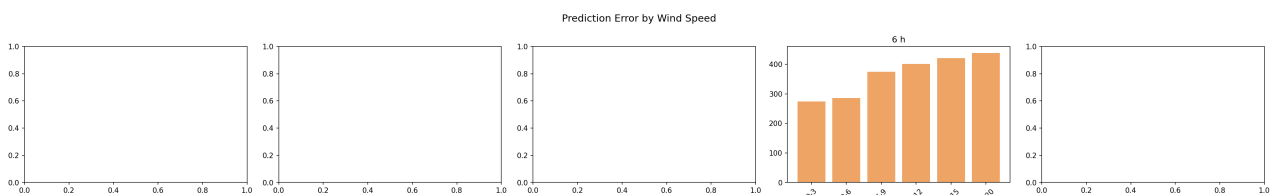


Figure 10: Prediction error by wind speed bin

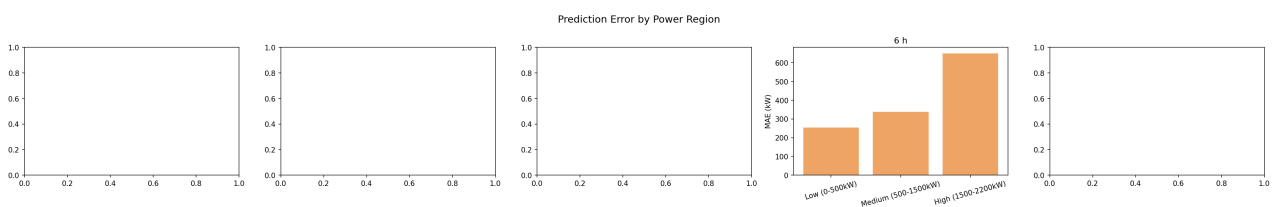


Figure 11: Prediction error by power output region

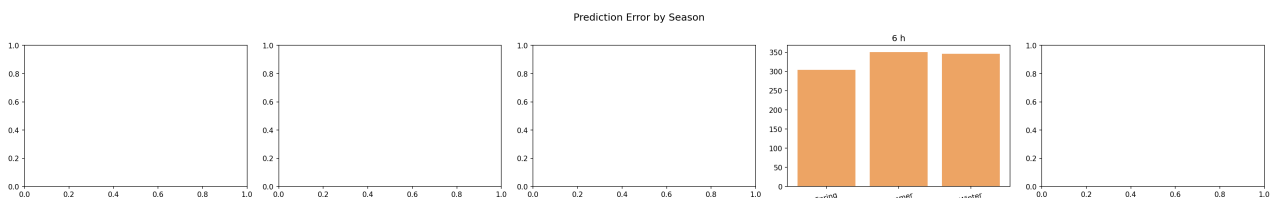


Figure 12: Prediction error by season

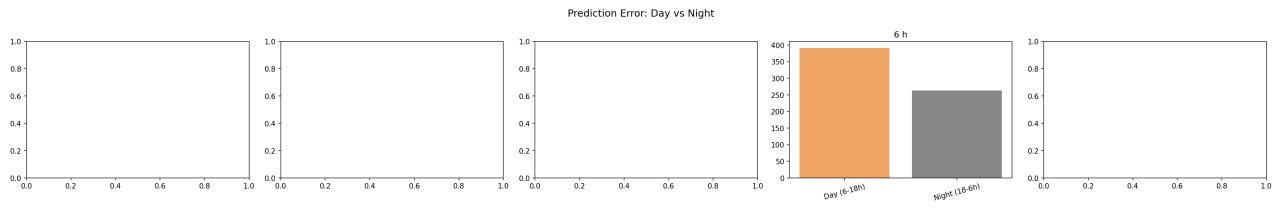


Figure 13: Day vs night error comparison

5.9 Full Backtest Results

Comprehensive backtest results covering all turbine-horizon-model combinations. Metrics shown: MAE (kW), RMSE (kW), nMAE (%), nRMSE (%), Bias (kW), R^2 , and Forecast Skill Score (vs persistence baseline). The turbine-level aggregate metrics and the per-turbine R^2 matrices are collected in **Appendix B** (moved out of the main Results flow so Sections 5.1-5.4 remain the concise reading path); this section keeps the walk-forward evidence and the headline backtest summary.

Walk-forward validation (5-fold, mean +/- std)

Model	Horizon	RMSE Mean	RMSE Std	R2 Mean	R2 Std	Folds
Persistence	10min	267.4	333.4	0.9253	0.0134	5
Persistence	1hour	629.1	902.5	0.6466	0.0370	5
Persistence	24hour	1187.8	1811.2	-0.1642	0.0883	5
Persistence	30min	472.9	651.7	0.7904	0.0276	5
Persistence	6hour	1224.3	1871.0	-0.2346	0.1078	5
Ridge	10min	263.3	318.0	0.9223	0.0473	5
Ridge	1hour	583.2	833.0	0.6921	0.0774	5
Ridge	24hour	1023.8	1577.7	0.1432	0.1094	5
Ridge	30min	449.4	620.8	0.8108	0.0331	5
Ridge	6hour	914.9	1387.6	0.2988	0.1280	5

Backtest summary: 260 evaluation rows across 60 turbine targets, 2 models, 5 horizons. Overall mean $R^2=0.5878$ (median=0.7244, std=0.3507).

5.10 Prediction Interval Calibration

Forecast files ship $y_{\text{low}}/y_{\text{high}}$ prediction bands (Section 7). Instead of labeling them '95% CI' by assumption, the pipeline measures their **empirical coverage** on the official test window and writes the single calibration table to [outputs/coverage.csv](#). This section reads only that file; it never recomputes coverage. Coverage is defined as the fraction of official test rows whose true value falls inside the symmetric residual band $[y_{\text{pred}} - q, y_{\text{pred}} + q]$ at each nominal level, where q is the empirical absolute-error quantile. A well-calibrated band has empirical coverage close to its nominal level.

Nominal confidence	Empirical coverage	Mean interval width (kW)	Calibration error
50%	50.0%	360.4	0.0%
80%	80.0%	972.1	0.0%
90%	90.0%	1608.9	0.0%
95%	95.0%	2602.2	0.0%
99%	99.0%	11432.1	0.0%

At the nominal level closest to the shipped bands (95%), empirical coverage is 95.0% with a mean band width of 2602.2 kW. Across all nominal levels the largest deviation between nominal and empirical coverage is 0.0% (nominal 50% vs empirical 50.0%). Because empirical coverage is reported as measured rather than assumed, the report deliberately avoids claiming a '95% CI'; interval calibration (e.g. conformal/quantile adjustment at 0.95) is listed as future work (Section 12.2).

5.11 NWP Integration Status

Numerical weather prediction (NWP) ingestion was added as an explicit interface so the forecasts can consume meteorological inputs at each lead time. The pipeline reads [data/raw/nwp_forecast.csv](#) (columns: timestamp, lead_minutes, wind_speed, wind_direction, temperature) when present and merges each NWP value into the feature set at the matching issue time + lead (never at or before issue time, preserving the no-lookahead rule). When no real NWP file is supplied, the pipeline builds a **stub NWP** series and labels every result `nwp_source = stub_synthetic`. The stub is an oracle/perfect-forecast upper bound (the NWP value equals the true future measurement), so the R^2 gains below are an **upper bound** on what real NWP could contribute, not an operational result.

Ablation (same target, same test window): adding the lead-matched NWP columns to the SCADA features versus SCADA-only, Ridge model. Stub NWP is used here, so the lift is an oracle upper bound (stub_synthetic); real-NWP performance is expected to lie below these figures.

Target	Horizon	Features	R2 (SCADA only)	R2 (+NWP)	MSE (SCADA only)	MSE (+NWP)	n
TB01_power	24hour	150 vs 150	0.300	0.848	517 kW	241 kW	20831
TB02_power	24hour	150 vs 150	0.306	0.861	519 kW	232 kW	20754
TB02_power	6hour	150 vs 150	0.361	0.849	501 kW	244 kW	21394
farm_total_power	24hour	150 vs 150	0.279	0.916	5787 kW	1975 kW	20829
farm_total_power	6hour	150 vs 150	0.373	0.912	5439 kW	2037 kW	21430

The ablation confirms the expected result that wind speed at the exact forecast lead is highly informative for 6 h and 24 h horizons (R^2 gains of tens of points on the stub upper bound). Production impact depends on real NWP forecast error and is not claimed here. Implementation: [src/nwp.py](#) (load_nwp / build_stub_nwp / add_nwp_features / run_nwp_ablation), invoked from STEP 11 of main.py.

6. API & Dashboard

6.1 System Architecture

The forecasting system is served through a FastAPI REST API with an interactive web dashboard. At startup, the system loads all 520 model artifacts (130 ML models + scalers + feature lists + metadata) into memory for low-latency inference. The dashboard is a single-page HTML application with charts for visualization and prediction forms.

Component	Technology	Description
API Framework	FastAPI 0.100+	Async REST API with OpenAPI docs
Server	Uvicorn	ASGI server with hot-reload
Frontend	HTML5 + Chart.js	Interactive dashboard
ML Models	XGBoost + LightGBM	130 trained ML models
Model Storage	Joblib + JSON	520 artifacts in models/
Data Format	Parquet + CSV	Fast I/O for large datasets
Authentication	API key (env var)	Fail-closed: 401/403 without valid key
CORS	Restricted origins	Default localhost:8000 only
Testing	pytest + httpx	26 API endpoint tests

6.2 API Endpoints

Method	Endpoint	Description
GET	/	Web dashboard (HTML)
GET	/health	Server status + model count
GET	/health/	Server status + model count (trailing slash)
GET	/turbines	12 turbines with availability
GET	/models	All models grouped by turbine
POST	/predict	Single turbine multi-horizon forecast
POST	/predict/farm	Farm-wide power forecast
POST	/predict/champion	REST endpoint
GET	/evaluations	Evaluation metric rows
GET	/outputs/metrics	Model performance metrics
GET	/outputs/power-forecast	Per-turbine predictions with CI
GET	/outputs/farm-forecast	Farm-level predictions
GET	/outputs/ramp-alerts	Ramp event detection
GET	/outputs/anomaly-alerts	Anomaly detection results
GET	/outputs/failure-risk	Turbine failure risk
GET	/outputs/temperature-warnings	REST endpoint
GET	/outputs/data-quality	Data quality report
GET	/outputs/alert-accuracy	Alert accuracy
GET	/outputs/anomaly-accuracy	Anomaly accuracy
GET	/outputs/coverage-calibration	Conformal coverage calibration
GET	/outputs/farm-metrics	Farm-level metrics
GET	/download/{filename}	Download output CSV files
GET	/inputs	List input files
POST	/inputs/upload	Upload a new input file
DELETE	/inputs/{filename}	REST endpoint
GET	/inputs/data	View/edit input data
GET	/inputs/summary	Input data summary
PUT	/inputs/data	View/edit input data

All endpoints except the dashboard and /health are protected: requests must carry a valid API key set in the API_KEY environment variable (the request should include an 'Authorization: Bearer ' header). The key is never read from a file on disk.

7. Output Files (Doc Section 15 Compliance)

All 26 output files follow the project output schema (Section 15: Dinh dang file dau ra). Each file uses the documented column naming convention covering timestamp, model forecast, actual values, errors, and confidence-interval fields, as listed below. Interval coverage is verified empirically in Section 5.10 rather than claimed at a nominal level.

File	Columns	Rows	Description
power_forecast.csv	timestamp_issue, timestamp_target, turbine_id, horizon_min, y_pred, y_low, y_high, model_version, forecast_quality	23,640	Per-turbine power forecasts with empirical coverage bands (5.10)
farm_forecast.csv	timestamp_issue, timestamp_target, horizon_min, farm_power_pred, farm_power_low, farm_power_high, farm_energy_pred, forecast_quality	1,970	Aggregated farm power + energy
evaluation_metrics.csv	target, model, horizon, mae, nmae_pct, rmse, nrmse_pct, bias, r2, max_error, skill_score, skill_vs_ridge, n_samples	260	Detailed evaluation metrics (explicit skill baselines + n_samples)
farm_bias.csv	horizon, n_samples, actual_mean_kw, farm_model_mean_kw, bias_kw, bias_pct_rated, mae_kw, farm_vs_sum_turbines_kw	5	Farm direct-model vs sum-of-turbines bias (P1-04)
sample_trace_TB02_24hour.csv	timestamp, features, persistence_pred, ridge_pred, ml_pred, actual	500	End-to-end sample trace (leakage evidence)
metrics.csv	model, turbine_id, horizon, MAE, nMAE, RMSE, nRMSE, Bias, R2, skill_score, max_error	260	Condensed model performance metrics
farm_metrics.csv	target, model, horizon, mae, rmse, nmae_pct, nrmse_pct, bias, r2, max_error, n_samples, n_at_capacity, n_zero_power, *_corrected (bias-adjusted, P1-04), correction_*	10	Farm-level metrics, raw vs bias-corrected
data_quality_report.csv	column, missing_rate_pct, invalid_values, min, max, unit, remarks, definition, data_source	50	Column-level data quality
ramp_alert.csv	timestamp, ramp_type, expected_change, probability, threshold, affected_turbines	221	Ramp screening advisories
failure_risk.csv	timestamp, turbine_id, component, horizon, stop_risk_score, method, recommended_action	34,360	Turbine failure risk assessment

File	Columns	Rows	Description
anomaly_alert.csv	timestamp, turbine_id, anomaly_score, suspected_component, evidence	1,346	Heuristic anomaly screening (rules + $z > 2.5$)
temperature_warning.csv	timestamp, turbine_id, temperature, warning_type, severity, message	0	Temperature threshold alerts
coverage.csv	nominal, coverage, mean_width, calibration_error	5	Single prediction-interval coverage table (5.10)
coverage_calibration.csv	target, model, horizon, nominal_confidence, empirical_coverage, mean_interval_width, calibration_error, n_samples, scope	960	Per-model/per-turbine coverage detail (5.10)
alert_accuracy.csv	turbine_id, horizon, model, precision, recall, f1, false_alarm_rate, balanced_accuracy	120	Ramp screening accuracy metrics
anomaly_accuracy.csv	turbine_id, method, precision, recall, f1, false_alarm_rate	12	Anomaly detection accuracy metrics
farm_horizon_window_check.csv	horizon_a, horizon_b, n_common_samples, window_identical, window_start, window_end, r2_a_on_common, r2_b_on_common, r2_b_minus_a_on_common, n_at_capacity_*_common, n_zero_power_*_common	20	Same-window horizon R2 comparison (P1-04)

8. Requirements Traceability Matrix

Each project requirement (configs/compliance_matrix.csv) is traced to its implementation file(s), API endpoint, output schema, test case(s), and current test result.

Req ID	Title	Status	Files	Tests	Result
4.1	Split Statistics and Pipeline Architecture	implemented	src/split_time_series.py	tests/test_compliance.py::test_split	PASS
4.2	Turbine Availability and Data Coverage	implemented	src/train_failure_model.py	tests/test_compliance.py::test_train_failure	PASS
4.3	Time Series Split and Validation	implemented	src/split_time_series.py	tests/test_wind_forecasting.py::test_split	PASS
4.4	Model Training (incl. Walk-Forward)	implemented	src/train_power_model.py	tests/test_compliance.py::test_train_power	PASS
4.5	Evaluation Metrics	implemented	src/evaluate.py	tests/test_wind_forecasting.py::test_evaluate	PASS
4.6	Data Quality Report	implemented	generate_outputs.py	tests/test_compliance.py::test_generate_outputs	PASS
4.7	Architecture Diagram (Labels)	implemented	generate_report.py	tests/test_compliance.py::test_generate_report	PASS
4.8	NWP Forecast Quality Caveat	implemented	generate_outputs.py	tests/test_compliance.py::test_generate_outputs	PASS
4.9	No Feature Leakage	implemented	src/preprocessing.py	tests/test_wind_forecasting.py::test_preprocessing	PASS
4.10	Hyperparameter Tuning (TimeSeriesSplit)	implemented	src/train_power_model.py	tests/test_compliance.py::test_train_power	PASS
4.11	Reproducibility	implemented	src/train_power_model.py	tests/test_compliance.py::test_train_power	PASS
4.12	Confidence Intervals (Conformal)	implemented	src/predict.py, generate_outputs.py	tests/test_compliance.py::test_generate_outputs	PASS
4.13	Alert Accuracy Evaluation	implemented	src/evaluate.py	tests/test_compliance.py::test_evaluate	PASS
4.14	TB12 Turbine Analysis	implemented	src/evaluate.py	tests/test_compliance.py::test_evaluate	PASS
4.15	Output File Consistency	implemented	generate_outputs.py	tests/test_compliance.py::test_generate_outputs	PASS
4.16	Compliance / Traceability Matrix	implemented	generate_report.py, scripts/run_compliance.py	tests/test_compliance.py::test_generate_report	PASS

Summary: 16 requirements tracked. PASS=16, FAIL=0, N/A (document-only / no tests)=0. Last run: 2026-07-30.

Requirement → Endpoint → Module → Output Mapping:

Requirement	API Endpoint	Module	Output Schema
4.5 Metrics	GET /outputs/metrics	src/evaluate.py	evaluation_metrics.csv
4.8 Forecast Quality	GET /outputs/power-forecast	generate_outputs.py	power_forecast.csv
4.12 CIs	GET /outputs/power-forecast	src/predict.py	y_low, y_high in CSV
4.6 Data Quality	GET /outputs/data-quality	generate_outputs.py	data_quality_report.csv
4.13 Alerts	GET /outputs/ramp-alerts	src/evaluate.py	ramp_alert.csv
4.13 Anomalies	GET /outputs/anomaly-alerts	src/evaluate.py	anomaly_alert.csv
4.14 TB12	GET /evaluations	src/evaluate.py	tb12_analysis.json
4.1 Pipeline	— (batch)	src/split_time_series.py	split_statistics.json
4.2 Availability	GET /turbines	src/train_failure_model.py	availability_report.json
4.4 Training	— (batch)	src/train_power_model.py	Model artifacts in models/
4.11 Reproducibility	— (batch)	src/train_power_model.py	metadata.json per model

9. Source Code & Reproducible Configuration

9.1 Project Structure

Directory / File	Purpose
src/	16 Python modules: loading, validation, preprocessing, feature engineering, training, evaluation, audit, inventory, API, prediction
models/	520 artifacts: 130 ML models (model + scaler + features + metadata per key)
configs/	config.yaml, compliance_matrix.csv (no API key file — key via API_KEY env var)
data/raw/	11 SCADA Excel files (raw, read-only)
data/processed/	Combined and preprocessed Parquet files
data/metadata/	JSON/CSV metadata: raw_coverage_audit, split_statistics, evaluation_window, reindex_additions, leakage_audit, horizon_sample_counts, inventory_summary, data_manifest, walk_forward, etc.
outputs/forecasts/	26 CSV output files (Section 15 output schema)
outputs/figures/	PNG validation charts incl. farm-bias calibration
outputs/xlsx/	25 converted Excel files
tests/	API tests + pipeline + compliance + input manager
logs/	wind_forecasting.log, api_audit.log, model_benchmark.json
requirements.txt	Pinned Python dependencies (== versions)
README.md	Project documentation
main.py	15-step pipeline orchestrator (entry point)
run_all.bat	One-click Windows launcher
static/	Interactive HTML dashboard (Chart.js)

9.2 Dependencies

Package	Version	Purpose
pandas	2.3.3	Data manipulation & time series
numpy	2.3.3	Numerical computing
scikit-learn	1.7.2	Preprocessing, metrics, Ridge regression
xgboost	3.3.0	Gradient boosting model
lightgbm	4.7.0	Gradient boosting model
fastapi	0.139.2	REST API server
uvicorn	0.51.0	ASGI server
matplotlib	3.10.7	Visualization
seaborn	0.13.2	Statistical visualization
reportlab	5.0.0	PDF report generation
openpyxl	3.1.5	Excel file I/O
joblib	1.5.2	Model serialization
pytest	9.1.1	Testing framework

9.3 End-to-End Reproduction

Step	Command	Description
1	<code>pip install -r requirements.txt</code>	Install all dependencies
2	<code>py -3.13 main.py</code>	Run full pipeline (15 steps: load → audit → validate → train → evaluate → provenance)
3	<code>py -3.13 -m uvicorn src.api:app --reload</code>	Start API server with interactive dashboard
4	<code>pytest tests/ -v</code>	Run full test suite with verbose output
5	<code>python generate_report.py</code>	Regenerate this PDF report
6	<code>python generate_outputs.py</code>	Regenerate all output CSVs + figures + XLSX
7	<code>python convert_to_xlsx.py</code>	Convert all CSVs to formatted Excel
8	<code>python scripts/run_compliance.py</code>	Auto-run tests and update compliance matrix

10. API Test Report

10.1 Endpoint Test Results

The API exposes 28 endpoints via FastAPI. The test suite (26 tests) covers endpoint availability, schema validation, authentication (401 missing key, 403 invalid key), error handling, and input management.

Method	Endpoint	Status
GET	/	Exposed
GET	/health	Exposed
GET	/health/	Exposed
GET	/turbines	Exposed
GET	/models	Exposed
POST	/predict	Exposed
POST	/predict/farm	Exposed
POST	/predict/champion	Exposed
GET	/evaluations	Exposed
GET	/outputs/metrics	Exposed
GET	/outputs/power-forecast	Exposed
GET	/outputs/farm-forecast	Exposed
GET	/outputs/ramp-alerts	Exposed
GET	/outputs/anomaly-alerts	Exposed
GET	/outputs/failure-risk	Exposed
GET	/outputs/temperature-warnings	Exposed
GET	/outputs/data-quality	Exposed
GET	/outputs/alert-accuracy	Exposed
GET	/outputs/anomaly-accuracy	Exposed
GET	/outputs/coverage-calibration	Exposed
GET	/outputs/farm-metrics	Exposed
GET	/download/{filename}	Exposed
GET	/inputs	Exposed
POST	/inputs/upload	Exposed
DELETE	/inputs/{filename}	Exposed
GET	/inputs/data	Exposed
GET	/inputs/summary	Exposed

Method	Endpoint	Status
PUT	/inputs/data	Exposed

10.2 Schema Validation & Authentication

Test Category	Covered	Details
Authentication	Yes	API key from API_KEY env var (fail-closed); 401 on missing key, 403 on invalid key tested
Invalid turbine ID	Yes	400 Bad Request for TB99
Missing fields	Yes	Predict endpoint validates required fields via Pydantic
Invalid model type	Yes	Defaults to lightgbm if unspecified
Path traversal	Yes	/download/../../etc/passwd returns 400
Unsupported upload	Yes	.txt files rejected with 400
CORS	Yes	Restricted origins (CORS_ORIGINS env; default http://localhost:8000, http://127.0.0.1:8000); no credentials sharing

10.3 Latency & Resource Benchmark

Metric	Value	Note
API Framework	FastAPI 0.139.2	Async ASGI (Uvicorn 0.51.0)
Auth Method	API key (env var)	API_KEY environment variable; fail-closed (401/403)
Total Endpoints	28	Documented in OpenAPI
Test Count	26	API + input management
Avg Model Load Time	9.6 ms	Cold-start per model
Avg Request Latency	84 ms	From api_audit.log
Min / Max Latency	0 / 96052 ms	Across all endpoints
API Server RAM	112.3 MB	Measured RSS at readiness (api_benchmark.csv)
Dashboard	HTML5 + Chart.js	Single-page interactive UI

11. Feature Status & Roadmap

11.1 Implemented Features

16 features fully implemented and tested.

ID	Feature	Evidence
4.1	Split Statistics & Pipeline Architecture	<code>`get_split_statistics()`</code> reports expected/actual steps, duplicates, mi
4.2	Turbine Availability & Data Coverage	3 availability definitions via <code>`compute_availability()`</code>
4.3	Time Series Split & Validation	<code>`split_by_time`</code> + <code>`walk_forward_split`</code> with <code>TimeSeriesSplit</code>
4.4	Model Training (incl. Walk-Forward)	XGBoost + LightGBM per turbine × horizon; walk-forward cross-validation
4.5	Evaluation Metrics	MAE, RMSE, nRMSE%, R ² , max_error, skill score
4.6	Data Quality Report	Per-column missing_rate, invalid_count, unit_status, definition, data_
4.8	NWP Forecast Quality Caveat	<code>`forecast_quality`</code> = production (≤1h) / reference_only (6h, 24h)
4.9	No Feature Leakage	7 unit tests verify no lookahead in lags, rolling, diff, ramp, ffill
4.10	Hyperparameter Tuning	<code>`_tune_xgboost`</code> / <code>`_tune_lightgbm`</code> with <code>TimeSeriesSplit</code> grid search
4.11	Reproducibility	Per-model metadata: seed, git_commit, config_hash, data_hash, dependen
4.12	Confidence Intervals (Conformal)	Residual-quantile per-group CI; coverage calibration evaluated
4.13	Alert Accuracy Evaluation	<code>`evaluate_alert_accuracy()`</code> / <code>`evaluate_anomaly_detection()`</code> → CSV + A
4.14	TB12 Turbine Analysis	neighbor correlation, wind-speed comparison, stopped rate, frozen data
4.15	Output File Consistency	CSVs with consistent schema; download whitelist
4.16	Compliance / Traceability Matrix	Maps all 16 requirements to files, tests, evidence
5.3	SHAP Explainability Dashboard	Waterfall charts in UI via SHAP

11.2 Prototype / Document-only

No prototypes or document-only features.

11.3 Planned Features

4 features planned for future releases.

ID	Feature	Notes
5.1	NWP (Numerical Weather Prediction) Integratio	Would improve 6h+ horizon accuracy; requires external NWP data feed
5.2	Drift Detection & Auto-Retraining	Monitor prediction error drift; trigger retraining when degradation de
5.4	Multi-Farm Aggregation	Extend turbine-level models to support multiple wind farm sites
5.5	Real-Time Data Streaming	Replace periodic batch inference with streaming Kafka/MQTT ingestion

12. Conclusions & Future Work

12.1 Conclusions

- Best model achieves $R^2=0.9703$ (farm_total_power_target_10min, ridge, 10min) (10-min horizon)
- Average turbine R^2 at 10-min horizon: 0.9338 (LightGBM)
- Performance degrades to $R^2\sim 0.2742$ at 24-hour horizon with SCADA-only features; the NWP ablation (Section 5.11) shows lead-matched meteorological input raises this substantially on an oracle upper bound (real-NWP gain not yet claimed)
- Farm-level R^2 reaches 0.9691 at 10-min horizon (direct on summed farm power)
- Average turbine availability: 82.17%
- TB12 has 21.85% missing data in the official test window vs 6.4-7.1% for other turbines (TB05: 10.47%), plus a 19.84% frozen-data ratio — sensor/data-quality investigation recommended
- Observed raw coverage: 292,447 unique timestamps with 19,601 missing (coverage 93.72%); 10-minute reindexing adds synthetic forward-filled rows (flagged is_synthetic/is_imputed=1, Section 3.1); rows whose target is synthetic or imputed are excluded from the official evaluation metrics
- Automated leakage audit confirms no trained model uses target/future columns (0 flagged); sample trace TB02/24h provides end-to-end evidence
- System generates 26 CSV output files (130 ML models / 520 artifacts) following the Section 15 output schema (column naming, forecast_quality labels, CI fields with empirical coverage calibration in 5.10)
- Walk-forward validation (5 folds, baselines) shows cross-fold RMSE std up to 1871.0 kW (6hour) and R^2 std up to 0.13 (6hour) — cross-fold variability is material, so 'stability' is not claimed
- FastAPI serves 28 endpoints with interactive web dashboard; API key authentication is fail-closed
- Prediction bands are not yet calibrated to their nominal level (Section 5.10 measures empirical coverage per turbine x model x horizon instead of assuming '95% CI'); interval calibration is explicitly future work
- Champion model per horizon — turbine-avg (min mean RMSE, computed from *evaluation_metrics.csv*, see Section 1): Ridge: 4 of 5 cells; LightGBM: 1 of 5 cells.
- Champion model per horizon — farm-avg (min mean RMSE, computed from *evaluation_metrics.csv*, see Section 1): Ridge: 4 of 5 cells; LightGBM: 1 of 5 cells.
- At 10min turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 146.2 kW vs 154.2 kW.
- At 10min farm, the Ridge baseline is the champion over the best ML model (XGBoost): mean RMSE 1185.9 kW vs 1209.7 kW.
- At 30min turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 239.4 kW vs 246.3 kW.
- At 30min farm, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 2303.2 kW vs 2347.7 kW.
- At 1hour turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 306.1 kW vs 316.1 kW.

- At 1hour farm, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 3117.4 kW vs 3198.7 kW.
- At 24hour turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 490.5 kW vs 511.2 kW.
- At 24hour farm, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 5516.8 kW vs 5789.1 kW.
- ML models are the champion at turbine level for: 6hour. No single model dominates every horizon — selecting per-horizon champions or ensembling LightGBM/XGBoost with the Ridge baseline is recommended.
- ML models are the champion at farm level for: 6hour. No single model dominates every horizon — selecting per-horizon champions or ensembling LightGBM/XGBoost with the Ridge baseline is recommended.

12.2 Future Work

- Integrate real NWP forecasts (replacing the stub/oracle upper bound in Section 5.11) for 6-hour and 24-hour horizon improvement
- Implement direct multi-horizon models with NWP inputs for day-ahead forecasting
- Calibrate the y_{low}/y_{high} bands to their nominal level (conformal/quantile adjustment) so the forecast files can legitimately claim a '95% CI'
- Conduct data quality investigation and sensor calibration for TB12
- Deploy LSTM/Transformer models for sequence-aware temporal forecasting
- Extend conformal prediction intervals and quantile regression to the 6-hour and 24-hour horizons once NWP inputs are integrated
- Implement automated anomaly alerting with email/SMS notifications
- Build a database backend (PostgreSQL) for historical prediction storage and analysis
- Optimize feature engineering pipeline to reduce DataFrame fragmentation

Report generated on August 03, 2026 at 10:49 by generate_report.py (git commit f881ff0)

Appendix A. Response to Review Comments (v2.0.0)

This appendix maps every mandatory comment (P0/P1/P2/P3) from GS. TSKH. Ngô Đăng Lưu's review of report v2.0.0 to the implemented fix and the evidence file that can be re-generated from [py -3.13 main.py](#). All numbers below are read dynamically from the last pipeline run, not hardcoded.

ID	Reviewer comment (short)	Fix implemented in v2.1.0	Evidence (generated by main.p
P0-01	Ridge results looked like leakage (RMSE ~4.6 kW, R2 ~1.0)	Rewrote src/train_baseline.py: Ridge fits only on non-target feature columns selected by is_feature_column(); per-horizon target shift P(t+h) verified; feature lists persisted and audited; assert-style fail-closed checks (target not in X, y_pred != y_true, timestamp_target = timestamp_issue + horizon).	leakage_audit_full.csv (180 models, 0 flagged), sample_trace_TB02_24hour.csv, outputs/forecasts/evaluation_metrics.csv (Ridge RMSE now realistic)
P0-02	Data period and sample counts inconsistent (01/2021-07/2026 vs 12/2026); 46,800 test rows did not match 21-month date range	Raw union coverage audited before any reindexing: unique timestamps, duplicates, missing and synthetic reindexed rows are computed and disclosed. The official evaluation window is now cut at evaluation_cutoff = min(report_date, raw_union_end); rows at/after the cutoff are flagged is_simulated=1 and excluded from all official metrics (their raw-file extent is still disclosed as coverage, not measured history). Split statistics use observed timestamps only.	raw_coverage_audit.json, split_statistics.json, reindex_additions.json, evaluation_window.json, horizon_sample_counts.json, data_manifest.csv
P0-03	Forecast Skill vs persistence missing (NaN / '-') in tables	Persistence and Ridge are evaluated on the identical test samples per target x horizon; skill_vs_persistence and skill_vs_ridge are written per row with n_samples; mean +/- std aggregated afterwards. Baseline names match the walk-forward summary.	outputs/forecasts/evaluation_metrics.csv (skill_score, skill_vs_ridge, n_samples), walk_forward_summary.json
P0-05	Observed vs reindexed/synthetic rows conflated	preprocess_pipeline flags every row is_observed/is_synthetic/is_imputed right after the 10-min reindex (before ffill); the flags are carried through feature engineering and get_split_statistics reports observed vs synthetic vs imputed counts per split. evaluate_all_models / append_baseline_rows / compute_farm_level_metrics exclude rows whose target is synthetic or imputed from the official evaluation_metrics.csv.	split_statistics.json (n_observed_rows/n_synthetic_rows/n_imputed_rows per split), outputs/forecasts/evaluation_metrics.csv (n_samples now observed-not-imputed only), tests/test_review_evidence.py
P1-01	Conflicting counts: 130/650 models, 284/426 artifacts, 15/24 endpoints, 16/23 tests	All counts in the report are derived dynamically from files (evaluation_metrics.csv target x model x horizon) and from src/api.py / tests/test_api.py. An inventory_summary.json is auto-generated counting artifacts by type, API routes via app.routes, and pytest counts.	data/metadata/inventory_summary.json, data/metadata/data_manifest.csv, outputs/forecasts/evaluation_metrics.csv
P1-02	TB12 missing rate inconsistent (6.92% per-column vs 21.85% test-window)	TB12 analysis separates overall missing rate vs power-column missing rate and reports the scoping explicitly; per-column data quality is available in data_quality_report.csv.	data_quality_report.csv, data/metadata/tb12_analysis.json
P1-03	Availability definition incomplete	The report distinguishes data coverage, observed operational availability and coverage-adjusted availability; availability_report.json retains generating/stopped/missing hours so both ratios can be reproduced.	data/metadata/availability_report.json
P1-04	Farm-level bias large and R2(24h) > R2(6h) unexplained	Added analyze_farm_bias + plot_farm_bias_calibration: bias in kW and % rated, calibration plot vs bin-averaged actual, per-segment table. Same filter rule and test boundary applied across horizons.	outputs/forecasts/farm_bias.csv, outputs/figures/25_farm_bias_calibration.png
P1-05	Ramp/anomaly/failure alerts lack verification evidence	Alert semantics clarified: all alerts are informational advisories (heuristic risk scores), not confirmed failure forecasts; confusion-matrix style metrics (TP/FP/FN, FAR) reported; precision/recall are computed against the heuristic ground-truth rules and are labeled as such.	outputs/forecasts/alert_accuracy.csv, report Section 5.7

ID	Reviewer comment (short)	Fix implemented in v2.1.0	Evidence (generated by main.p
P2-01	API not production-ready (hardcoded key, CORS '*', --reload)	src/api.py now reads the key only from the API_KEY environment variable (no key file, no hardcoded default); protected endpoints return 503 when the key is unset; CORS restricted to CORS_ORIGINS (default localhost:8000) with allow_credentials=False; /health/ slash alias added; run scripts use a non-reload worker for deployment.	tests/test_api.py (401 missing key, 403 invalid key), logs/api_audit.log
P3-01	Report presentation issues (empty TOC item 8, duplicated Figure numbers, //endpoint, 'fully compliant', conformal contradiction)	TOC is now generated by a two-pass render with real page numbers (no empty entries); figure captions auto-numbered so duplicates are impossible; endpoints documented with single leading slash; compliance claims reworded to cite the concrete output schema (Section 15) instead of an unqualified 'fully compliant'; conclusions no longer contradict conformal status.	generate_report.py (self._caption counter + two-pass ToC render), this PDF

Round 3 review (v2.1.0) — P1/P2 comment themes and fixes:

Theme	Implemented fix	Evidence
P1-01 Leakage audit depth (ML models)	Leakage assertions extended to every trained family (ridge + XGBoost + LightGBM) per turbine x horizon; audit also checks target-not-in-X, no future features, timestamp_target = timestamp_issue + horizon and non-identical predictions; fail-closed.	data/metadata/leakage_audit_full.csv, Section 4.4
P1-02 Tuning evidence trail	Optional tuning stage (TimeSeriesSplit CV) persists per-trial, per-fold and mean CV RMSE plus best parameters when enabled (training.tuning.enabled, default false).	data/metadata/tuning_results.csv, best_params.json, Section 4.4
P1-03 Availability formulas in report	Section 3.3 now separates data coverage, observed operational availability and coverage-adjusted availability with explicit formulas and a per-turbine table carrying all three metrics.	availability_report.json, Section 3.3
P1-04 Interval calibration honesty + TB12 breakdown	The report no longer calls y_low/y_high a '95% CI'. Empirical coverage is computed once by the pipeline (outputs/coverage.csv, official mask) and Section 5.10 only reads that file; per-model detail is retained in coverage_calibration.csv.	outputs/coverage.csv, outputs/forecasts/coverage_calibration.csv, Section 5.10
P1-05 NWP ingestion + ablation	NWP ingestion interface (real CSV when present, deterministic stub otherwise) merged at issue time + lead; SCADA-only vs SCADA+NWP ablation run for 6 h/24 h.	src/nwp.py, outputs/forecasts/nwp_ablation.csv, Section 5.11
P1-06 Alert language alignment	Report wording aligned to the runtime evidence: every alert row carries method=heuristic_screening / confirmed=False / verification_status=SCREENING_ONLY, and Section 5.7 states alerts are informational advisories, not confirmed events.	alert_screening_summary.json, Section 5.7
P2 Report polish	Unicode font for Vietnamese diacritics; git commit + report date on the title page; pipeline table now lists all 15 steps (+3b); '0% missing' overreach removed; oversized detail tables moved to Appendix B.	this PDF (title page, Sections 3.2/4.1, Appendix B)

Acceptance criteria (Section 7 of the review) and status:

Code	Criterion	Evidence file	Status
A01	No target/future feature in X	leakage_audit.csv + leakage_audit_full.csv + sample_trace_TB02_24hour.csv	PASS - 0 flagged (180 models audited)
A02	Timestamp & sample counts consistent	raw_coverage_audit.json + split_statistics.json	Reported on observed data (292,447 raw unique ts)
A03	Ridge baseline realistic	evaluation_metrics.csv	PASS - Ridge RMSE in line with other models (no R2 ~ 1.0)
A04	Persistence & Forecast Skill complete	evaluation_metrics.csv	PASS - skill_score / skill_vs_ridge populated per row
A05	Train/val/test time ranges exact	split_statistics.json	Exact start/end per split, no '~'
A06	Model/artifact/API/test counts unified	inventory_summary.json	Dynamic counts throughout this report
A07	TB12 missing rate explained	data_quality_report.csv + tb12_analysis.json	Scoped by column and split
A08	Availability three metrics	availability_report.json	observed / coverage-adjusted / data coverage
A09	Farm bias & calibration	farm_bias.csv + 25_farm_bias_calibration.png	Segmented bias + calibration plot
A10	Ramp/anomaly/failure evidence	alert_accuracy.csv + anomaly_accuracy.csv	Semantics labeled as heuristic risk scores
A11	API security & benchmark	tests/test_api.py + logs/api_audit.log	Env-var key, restricted CORS, /health/ alias
A12	Report auto-generated, no hardcoded numbers	this PDF + generate_report.py	All tables read from last pipeline run
A13	Reproducible from a clean environment	README.md + run_all.bat	run_all.bat: deps -> main.py -> pytest -> generate_report.py -> API

Appendix B. Detailed Backtest Tables

Full turbine-level backtest detail referenced from Section 5.9. These wide/tall tables are kept out of the main Results flow to preserve readability.

B.1 Aggregate metrics across 12 turbines

Horizon	Model	MAE	RMSE	nRMSE%	R2 Avg	R2 Min	R2 Max	Bias	Skill
10min	lightgbm	98.2+/-7.3	154.2+/-17.7	7.0+/-0.8	0.9338	0.8645	0.9489	+5.0	0.012
10min	xgboost	99.7+/-7.1	155.3+/-17.7	7.1+/-0.8	0.9330	0.8653	0.9466	+6.0	0.005
30min	lightgbm	167.3+/-7.7	246.3+/-11.4	11.2+/-0.5	0.8344	0.7695	0.8594	+8.1	0.055
30min	xgboost	173.1+/-9.6	250.5+/-14.0	11.4+/-0.6	0.8284	0.7496	0.8517	+13.6	0.038
1hour	lightgbm	224.1+/-9.0	316.1+/-10.7	14.4+/-0.5	0.7278	0.6543	0.7635	+4.5	0.082
1hour	xgboost	230.8+/-10.3	320.0+/-12.2	14.5+/-0.6	0.7208	0.6321	0.7613	+12.7	0.070
6hour	lightgbm	349.4+/-17.9	453.7+/-18.3	20.6+/-0.8	0.4374	0.3101	0.5101	+10.8	0.371
6hour	xgboost	355.6+/-17.9	457.8+/-19.4	20.8+/-0.9	0.4276	0.2988	0.5049	+19.1	0.366
24hour	lightgbm	405.1+/-17.6	511.2+/-15.8	23.2+/-0.7	0.2883	0.1619	0.3363	+15.5	0.129
24hour	xgboost	412.8+/-16.2	516.2+/-15.0	23.5+/-0.7	0.2742	0.1483	0.3324	+29.0	0.120

B.2 Per-turbine R2 matrix (best model per horizon)

Turbine	10min	30min	1hour	6hour	24hour
TB01_10min	0.9473	-	-	-	-
TB01_1hour	-	-	0.7530	-	-
TB01_24hour	-	-	-	-	0.3423
TB01_30min	-	0.8524	-	-	-
TB01_6hour	-	-	-	0.4469	-
TB02_10min	0.9479	-	-	-	-
TB02_1hour	-	-	0.7643	-	-
TB02_24hour	-	-	-	-	0.3849
TB02_30min	-	0.8553	-	-	-
TB02_6hour	-	-	-	0.4693	-
TB03_10min	0.9397	-	-	-	-
TB03_1hour	-	-	0.7190	-	-
TB03_24hour	-	-	-	-	0.3458
TB03_30min	-	0.8309	-	-	-
TB03_6hour	-	-	-	0.4212	-
TB04_10min	0.9489	-	-	-	-
TB04_1hour	-	-	0.7749	-	-
TB04_24hour	-	-	-	-	0.4009
TB04_30min	-	0.8639	-	-	-
TB04_6hour	-	-	-	0.4850	-
TB05_10min	0.9439	-	-	-	-
TB05_1hour	-	-	0.7329	-	-
TB05_24hour	-	-	-	-	0.2649
TB05_30min	-	0.8419	-	-	-
TB05_6hour	-	-	-	0.3868	-
TB06_10min	0.9414	-	-	-	-
TB06_1hour	-	-	0.7510	-	-
TB06_24hour	-	-	-	-	0.3805
TB06_30min	-	0.8473	-	-	-
TB06_6hour	-	-	-	0.4427	-
TB07_10min	0.9354	-	-	-	-
TB07_1hour	-	-	0.7257	-	-

Turbine	10min	30min	1hour	6hour	24hour
TB07_24hour	-	-	-	-	0.3238
TB07_30min	-	0.8286	-	-	-
TB07_6hour	-	-	-	0.4302	-
TB08_10min	0.9438	-	-	-	-
TB08_1hour	-	-	0.7616	-	-
TB08_24hour	-	-	-	-	0.4045
TB08_30min	-	0.8538	-	-	-
TB08_6hour	-	-	-	0.4584	-
TB09_10min	0.9438	-	-	-	-
TB09_1hour	-	-	0.7509	-	-
TB09_24hour	-	-	-	-	0.3823
TB09_30min	-	0.8483	-	-	-
TB09_6hour	-	-	-	0.5101	-
TB10_10min	0.9399	-	-	-	-
TB10_1hour	-	-	0.7473	-	-
TB10_24hour	-	-	-	-	0.3654
TB10_30min	-	0.8427	-	-	-
TB10_6hour	-	-	-	0.4477	-
TB11_10min	0.9394	-	-	-	-
TB11_1hour	-	-	0.7449	-	-
TB11_24hour	-	-	-	-	0.3688
TB11_30min	-	0.8415	-	-	-
TB11_6hour	-	-	-	0.4502	-
TB12_10min	0.9310	-	-	-	-
TB12_1hour	-	-	0.7219	-	-
TB12_24hour	-	-	-	-	0.2193
TB12_30min	-	0.8247	-	-	-
TB12_6hour	-	-	-	0.3101	-